# TDM Fall 2024 John Deere Parts Demand Forecasting Class

## Point Forecasting Final Presentation

11/22/2024

Purdue University | College of Engineering

# *Agenda*

We will be going over our…

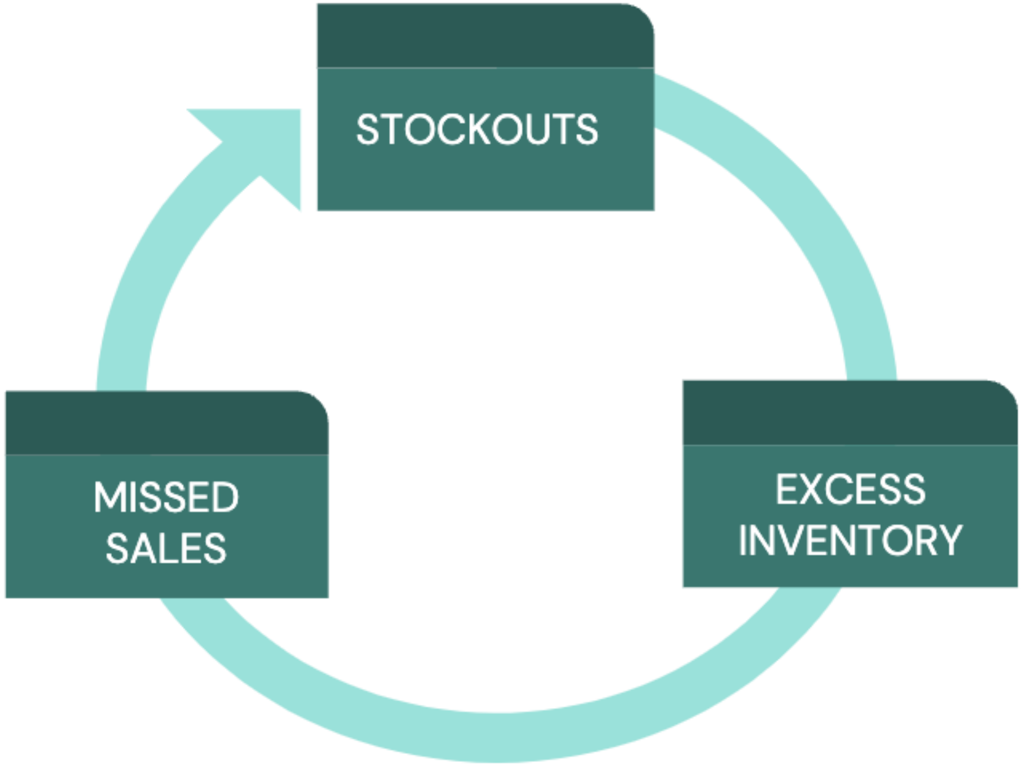
# *Business Problem*

## What do we need to do?

John Deere has a vast distribution of **1.6 million** part-location combinations. Due to the variability in demand patterns across different regions and product categories inaccurate forecasts can lead to **stockouts, excess inventory, and missed sales opportunities.**

This lack of demand visibility not only impacts the company's ability to maintain optimal inventory levels but also affects **customer satisfaction and operational efficiency.**

Thus, our objective is to deliver a precise 12-month demand forecast for a part-location time series, minimizing overprediction and addressing any bias, to enhance inventory management.



# *Project Scope*

## Business Problem

Limited External Data Integration

Ineffective Trend and Seasonality Capture

Suboptimal Traditional Model Performance

## Knowledge Gap

Lack of External Factors (Interest Rate, Weather)

Ineffective in capturing trend/seasonality effects

Traditional vs. Machine Learning Models

## Deliverables & Value Proposition

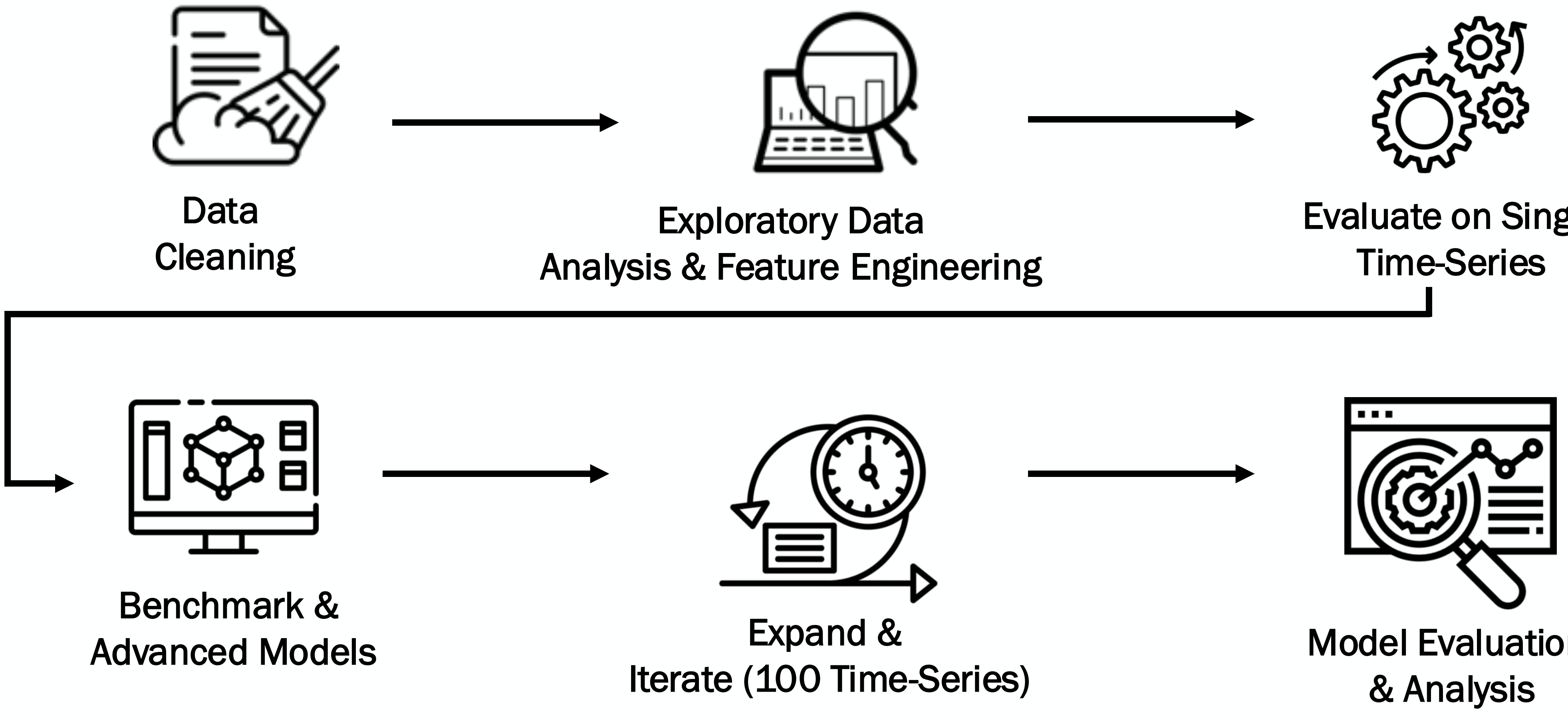
Integrate external market factors into the models

Perform EDA, Trend & Seasonality Analysis

Benchmark & compare with ML Models (Metrics)

# *Methodology*

## And workflow…



# Internal Dataset

## Key Takeaways...

**Part:** Unique identifier value for the John Deere part in question

**Location:** Unique identifier value for the dealership location where the part is needed or stored

**Date:** The "date" of recorded demand (**translates to "month" as the data was gathered monthly**), stored as a Date-Time format

**Demand Quantity:** Indicates the quantity of the part required at the specified location and date

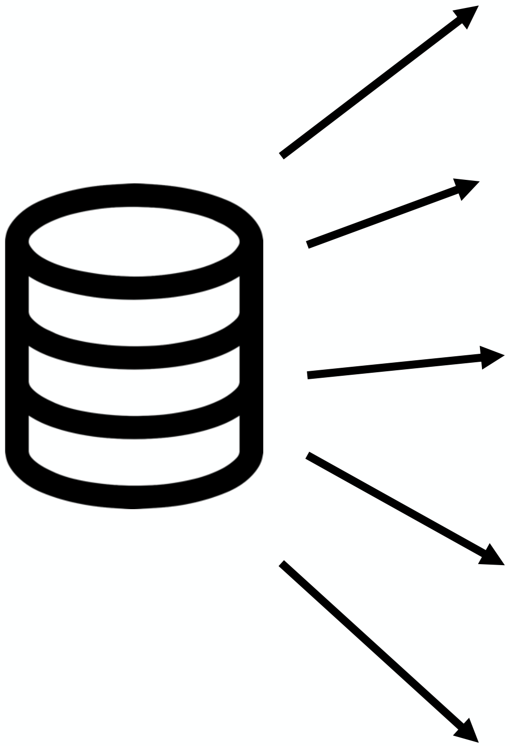
| Part | Location | Date | Quantity |
|---|---|---|---|
| 1371 | 28803 | 01-06-2021 | 102 |
| 1371 | 28803 | 01-03-2022 | 232 |
| 1371 | 28803 | 01-10-2022 | 94 |
| 1371 | 28803 | 01-01-2021 | 122 |
| 1371 | 28803 | 01-06-2022 | 124 |
| 1371 | 28803 | 01-05-2022 | 146 |
| 1371 | 28803 | 01-10-2021 | 96 |
| 1371 | 28803 | 01-04-2020 | 7 |
| 1371 | 28803 | 01-03-2023 | 258 |
| 1371 | 28803 | 01-04-2021 | 261 |
| 1371 | 28803 | 01-02-2021 | 80 |
| 1371 | 28803 | 01-07-2022 | 83 |
| 1371 | 28803 | 01-09-2022 | 88 |
| 1371 | 28803 | 01-08-2020 | 1 |
| 1371 | 28803 | 01-11-2020 | 100 |
| 1371 | 28803 | 01-10-2020 | 2 |
| 1371 | 28803 | 01-04-2022 | 293 |
| 1371 | 28803 | 01-01-2022 | 102 |
| 1371 | 28803 | 01-05-2020 | 0 |
| 1371 | 28803 | 01-02-2022 | 94 |
| 1371 | 28803 | 01-05-2021 | 151 |
| 1371 | 28803 | 01-01-2023 | 128 |
| 1371 | 28803 | 01-11-2022 | 94 |
| 1371 | 28803 | 01-08-2021 | 82 |
| 1371 | 28803 | 01-02-2023 | 133 |
| 1371 | 28803 | 01-12-2022 | 90 |

The dataset analyzed focuses on a single part and location, as indicated by the constant values in the "Part" and "Location" columns.

The "Quantity" column shows variability, with values ranging from 0 to 293, indicating fluctuations in demand over time (Normalization is important when considering more than 1 timeseries)

# External Dataset

## Key Takeaways…



**Rainfall:** Average Rainfall per Month across the United States

**Temperature:** Average Temperature per Month across the United States

**Federal Funds Rate:** The monthly federal funds rate (directly influences interest rates, buying, lending)

**Seasonality:** Observed patterns or fluctuations based on a month-based cycle.

**Trend**: Long-term movement or direction in data over an extended period

| | Quantity | Year | Month | Seasonal | FEDFUNDS | Rain Value |
|---|---|---|---|---|---|---|
| **0** | 5 | 2020 | 3 | -32.192708 | 0.65 | 34.81 |
| **1** | 7 | 2020 | 4 | -80.151042 | 0.05 | 34.12 |
| **2** | 0 | 2020 | 5 | -1.317708 | 0.05 | 32.74 |
| **3** | 4 | 2020 | 6 | 86.640625 | 0.08 | 32.10 |
| **4** | 1 | 2020 | 7 | -81.796875 | 0.09 | 32.28 |

The dataset uses Federal Funds Rate, Temperature, and Rainfall which are time-dependent variables

For future predictions, online forecast models will be used as input to the demand forecasting models

# *Data Preprocessing*

**Why and how?**

## Data Preprocessing Framework

- **Data Cleaning & Quality:** Remove missing, inconsistent, or outlier data to ensure accuracy and completeness.
- **Feature Engineering:** Create and merge meaningful time-dependent features such as seasonality, trends, & external factors.
- **Identify Part-Location Combination:** Group and find subset given unique part and location identifier.
- **Time Series Decomposition:** Break down the time series data into trend, seasonality, month, year and residual components.

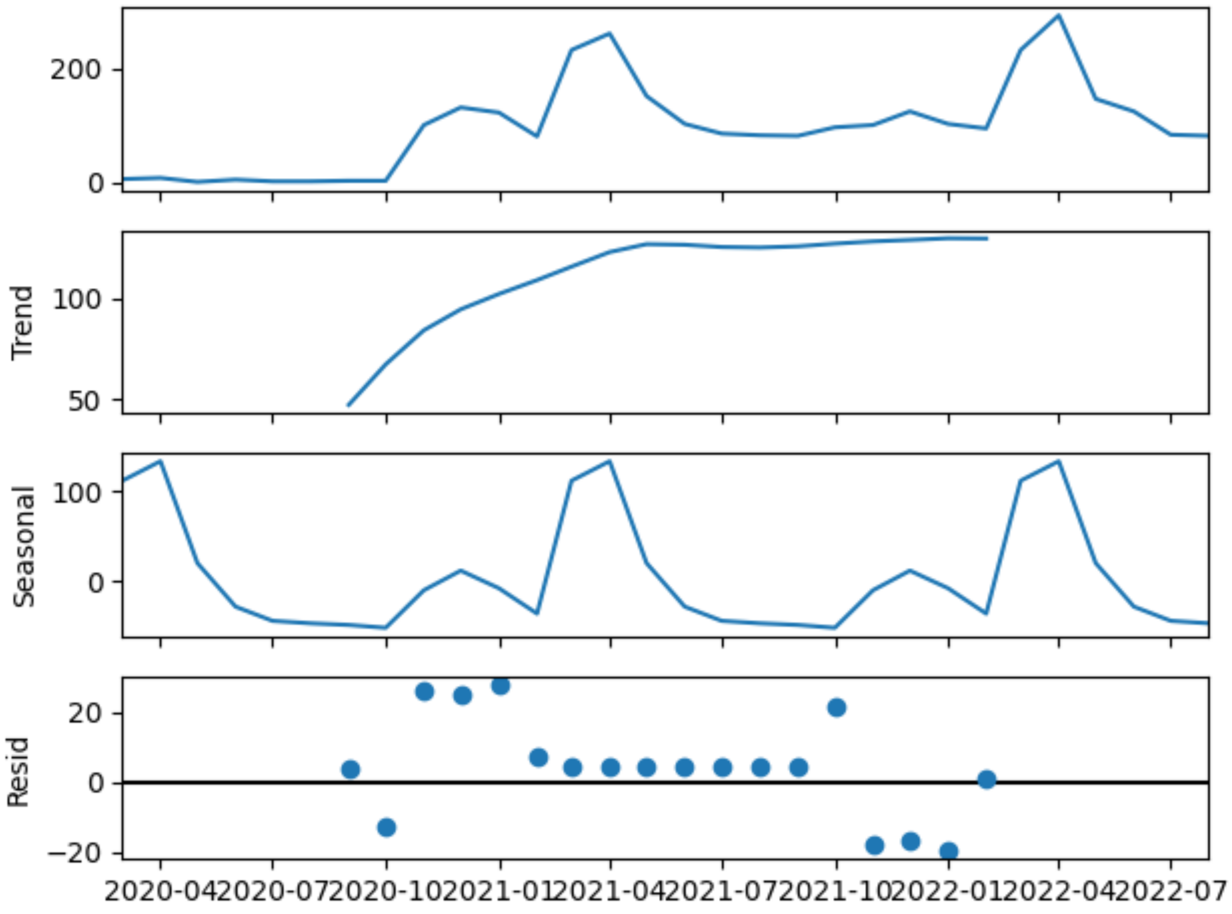
→ Layered Approach →

- **One-Hot Encoding:** Convert time-based categorical data to avoid month 12 vs month 1 bias
- **Merge External Factors:** Combine Federal Funds, Avg Temperature, and Avg Rainfall Time Based Data
- **Train/Test - Split:** For **Single Time Series**, use 80% of the dataset to train model, and test of 20% across models
- **Scaling & Normalization:** When extending to 100 timeseries, Qty Demanded affects RMSE value

# *Exploratory Data Analysis*

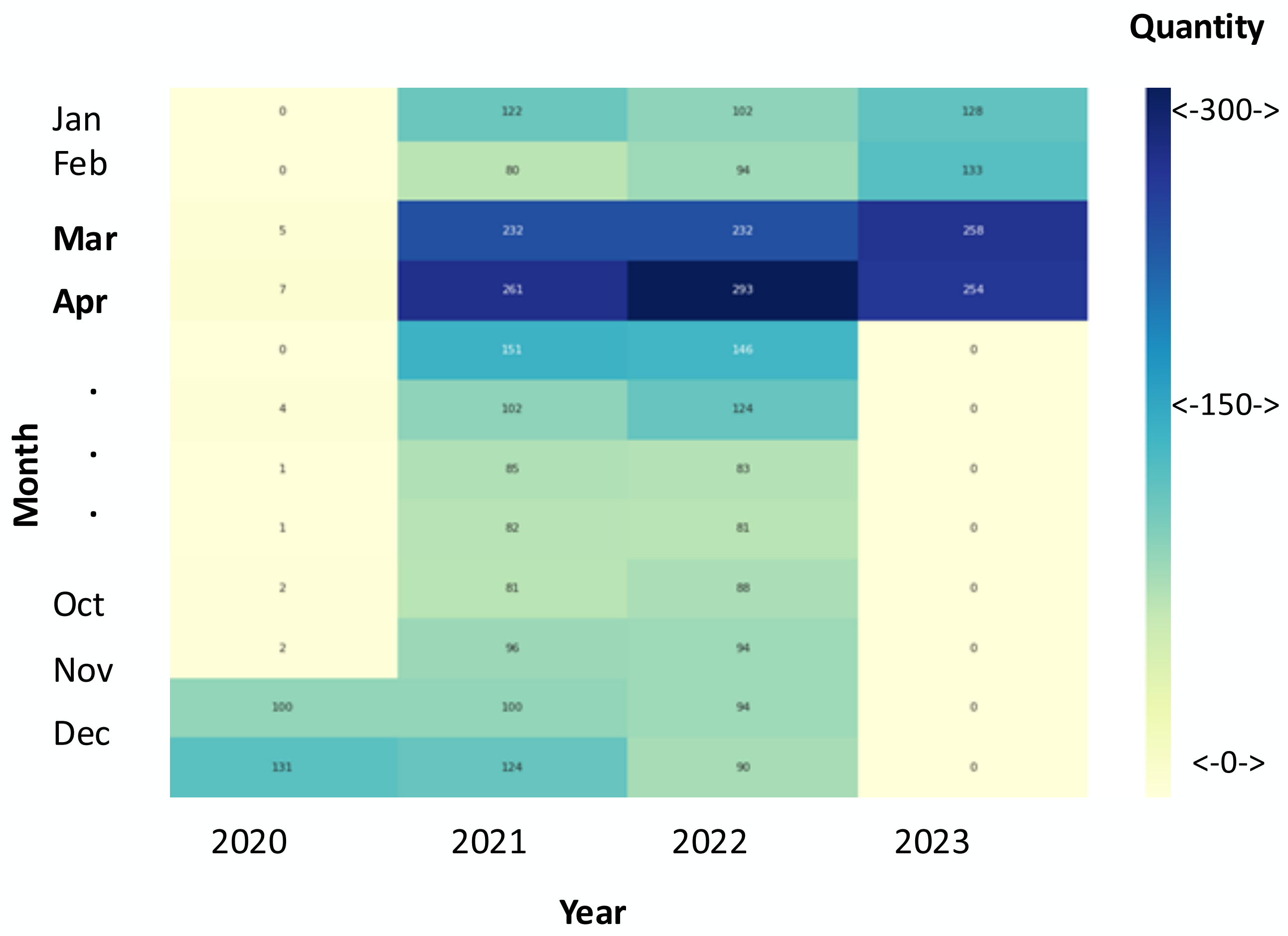
(EDA)

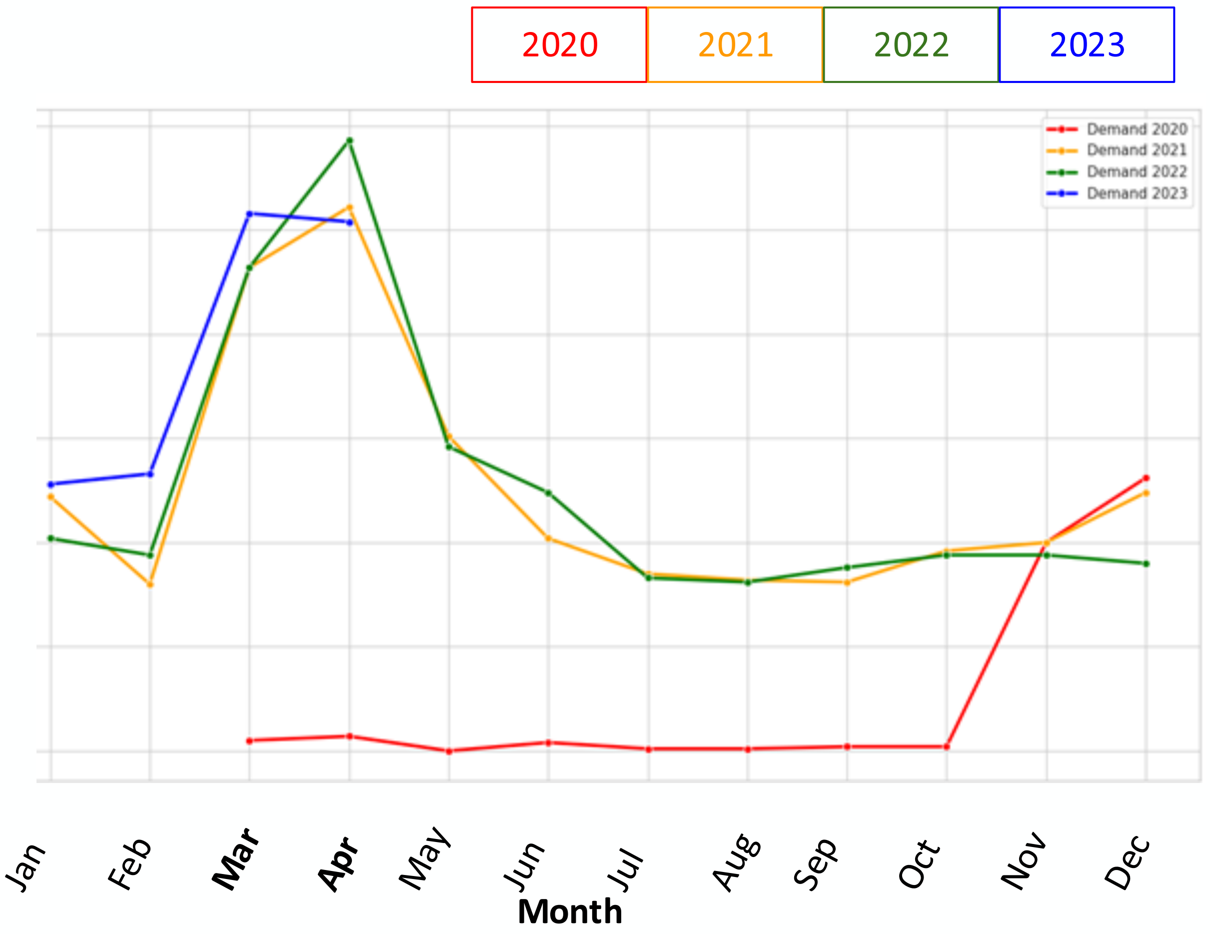
- **Trend and Anomaly Detection:** Identify and analyze current trends, patterns, and outliers in the data.
- **Correlation Analysis:** Examine relationships between key variables such as seasonality and trend influence the models, helping identify potential predictors for the model.
- **Seasonality and Variability:** Investigate recurring patterns (seasonality) and variability within the data, providing insight into predictable cycles and fluctuations.
- **Impact of External Variables:** Use correlation matrices and scatter plots to assess how external factors (e.g., weather, economic indicators) affect demand, highlighting their potential influence on model accuracy. .



# *Substantial Seasonal Peaks*

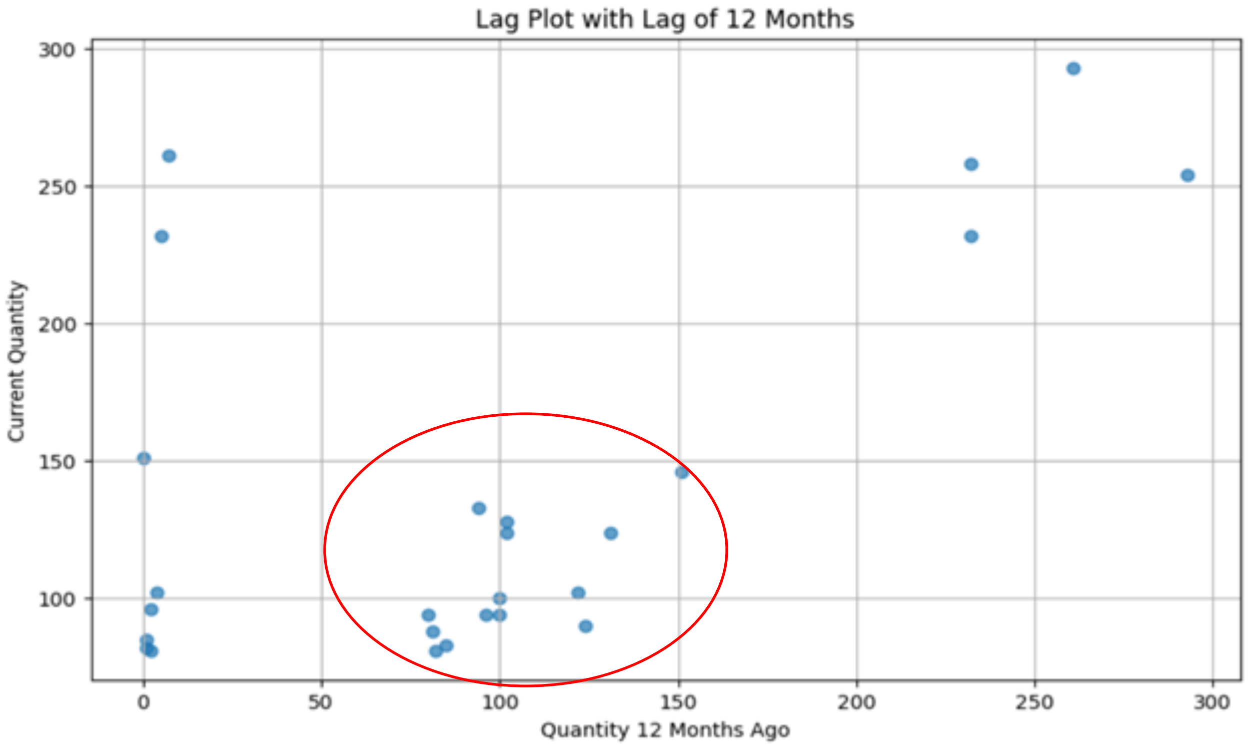
In both April and March.

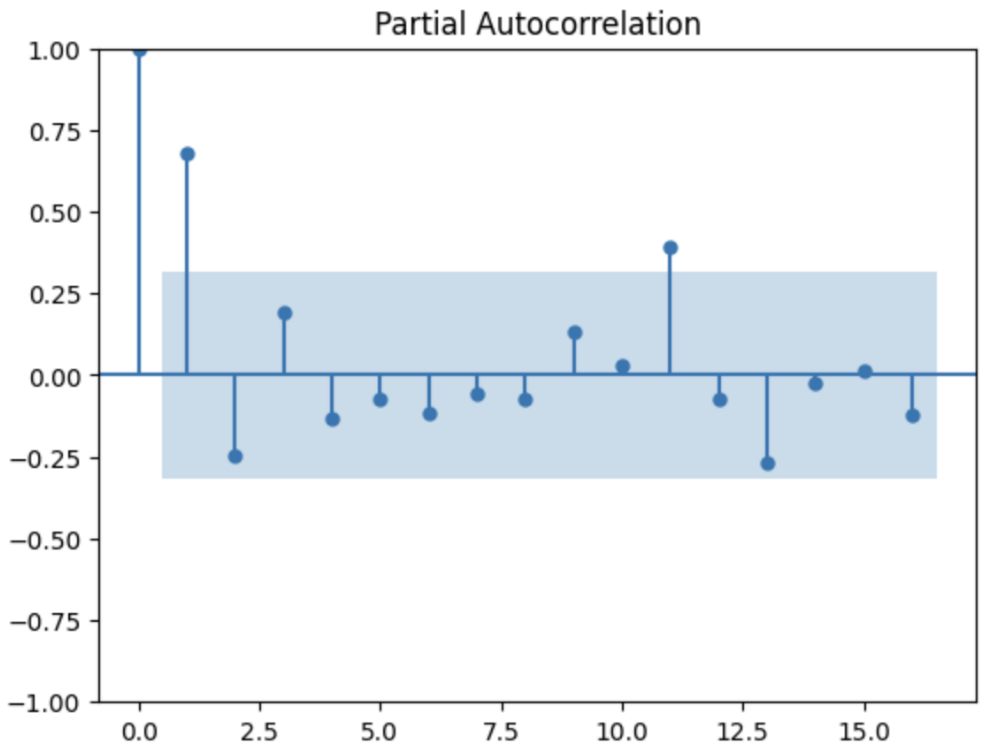


# *Lag plots confirm this seasonality,*

yet the data shows weak autocorrelation.



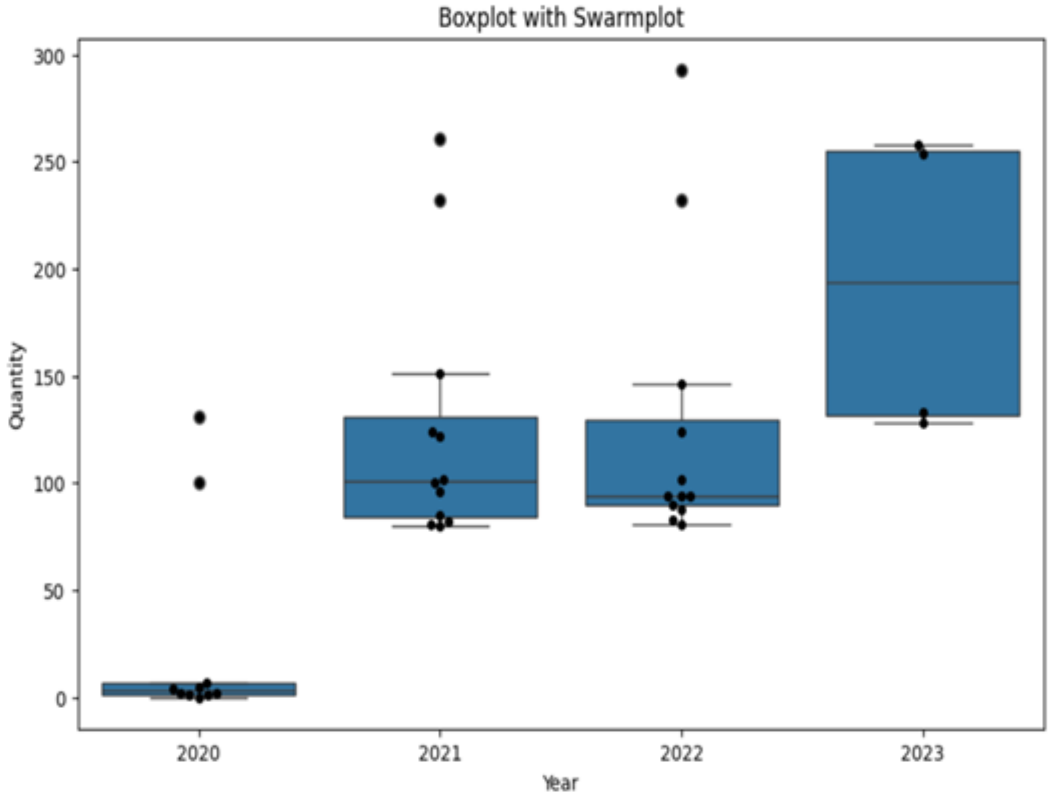



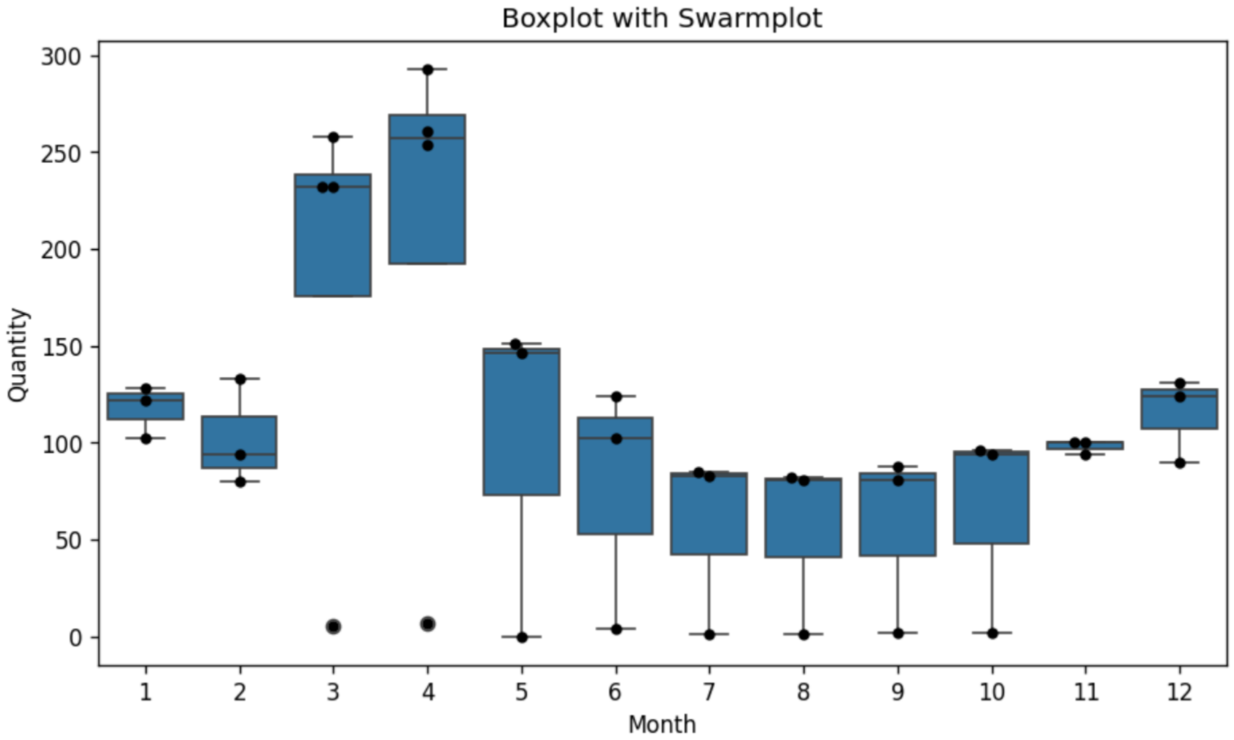

Clusters in the 12-month lag plot further demonstrate the seasonality in the dataset.

Shows correlation between lagged values within timeseries

# *This has to do with high variability*

Form month to month and year to year.



# *From Our EDA...*

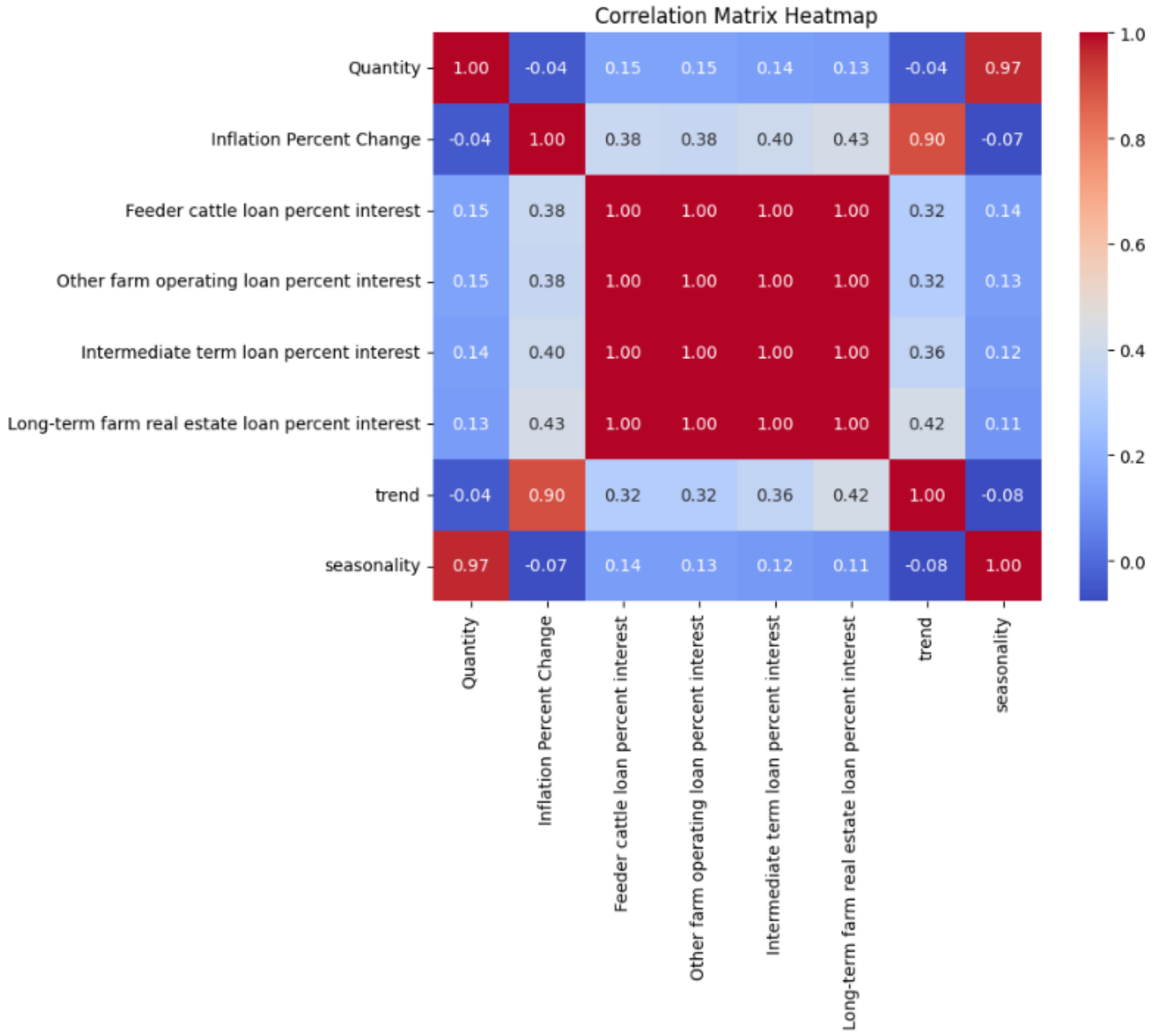
We found that:

We are working with a **seasonal dataset exhibiting high annual variability**;  thus, we concluded that incorporating external features may enhance our modeling.

# *So, what features should we incorporate?*

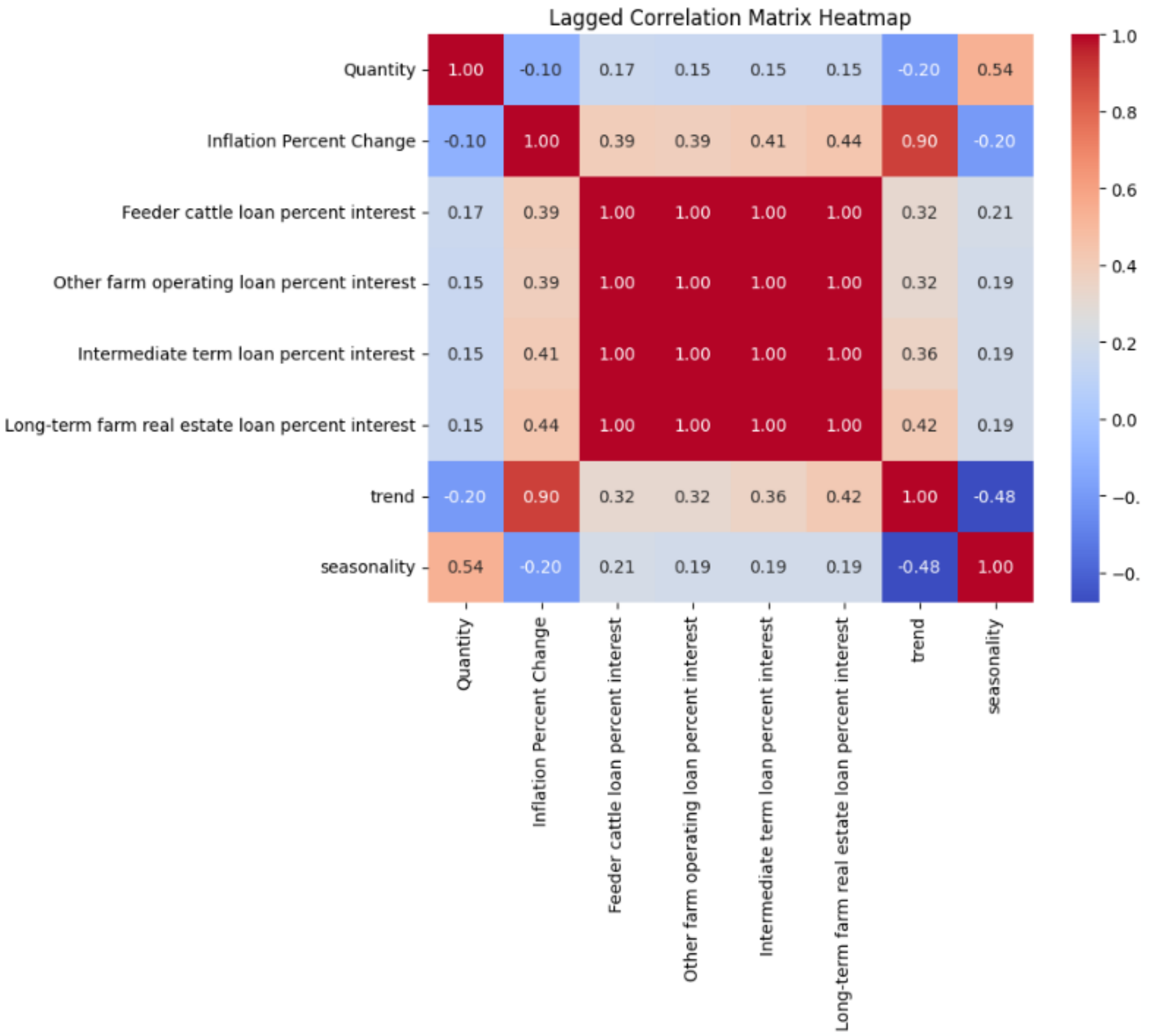
Let's look at a Linear Correlation Heatmap!

- Features initially investigated:
  - Inflation Rate Percent Change
  - Agricultural Loan interest Rates
  - Trend
  - Seasonality
- 1.00 (perfect) linear correlation across every Agriculture Loan Interest feature
  - This **data was only seasonal**, so the **same exact data was copied over** multiple times to achieve "monthly" data
- Seasonality and Demand Quantity has the highest linear correlation of 0.97
- Inflation Rate and Trend have linear correlation of 0.9
  - **We may be able to drop the supplemental Inflation Rate data**, as it overlaps with Trend



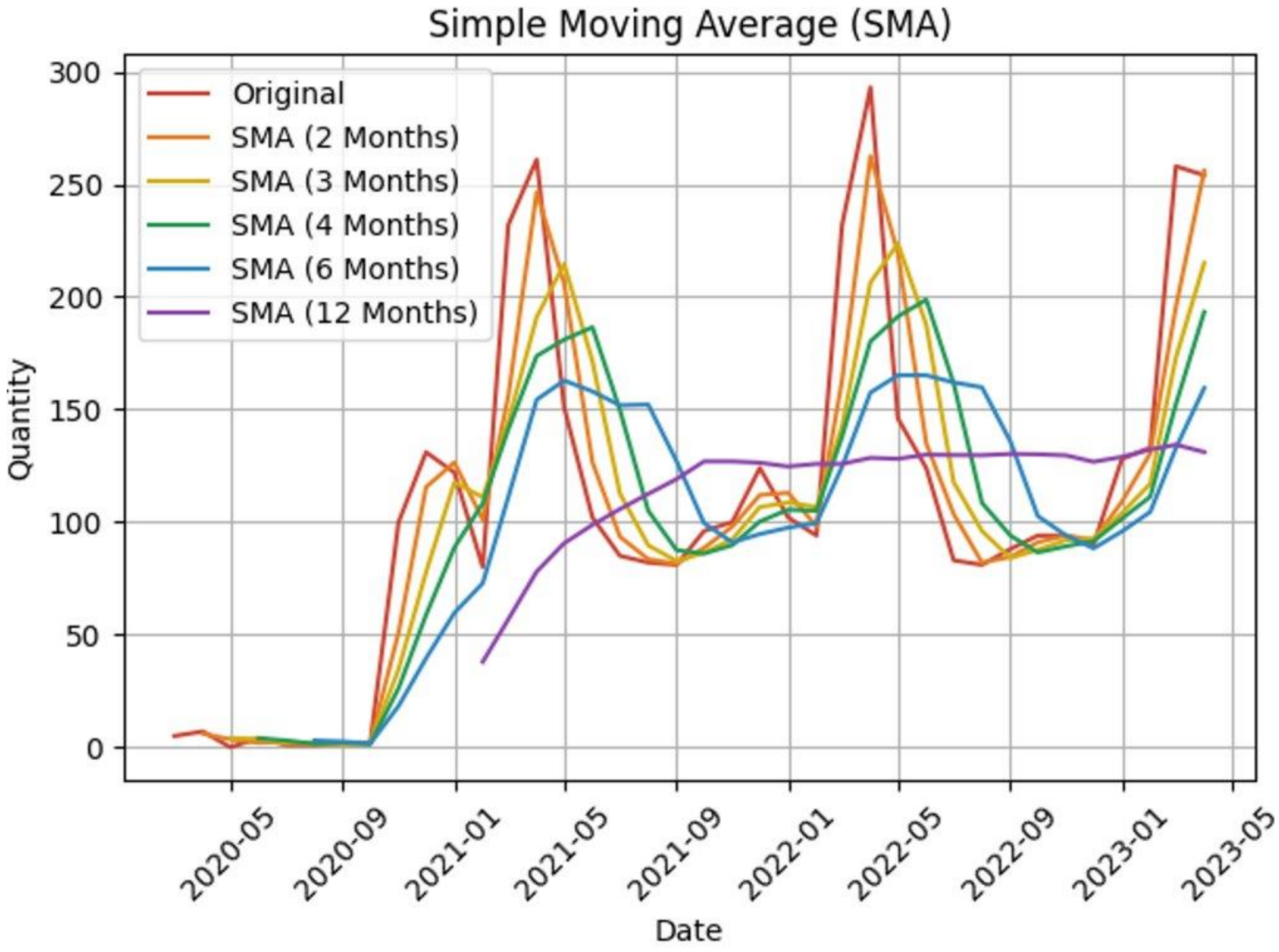
# *Lagged Heatmap*

- Values are **lagged by 1-month**,
  - Current month's data will correlate to next month's demand to **check if the current month's features could be predictors for the next month's demand.**
- Linear Correlation between Loan Interest Rates, Inflation Rate and Demand Quantity goes up slightly compared to Un-lagged (+0.01/0.02)
- Linear Correlation between Seasonality and Demand Quantity goes significantly down (-0.43)
- Moving forward, it may be wise to **lag the external features** (Loan Interest), while **keeping internal features** (Seasonality) **un-lagged**



# *Starting with Simple Moving Average (SMA)*

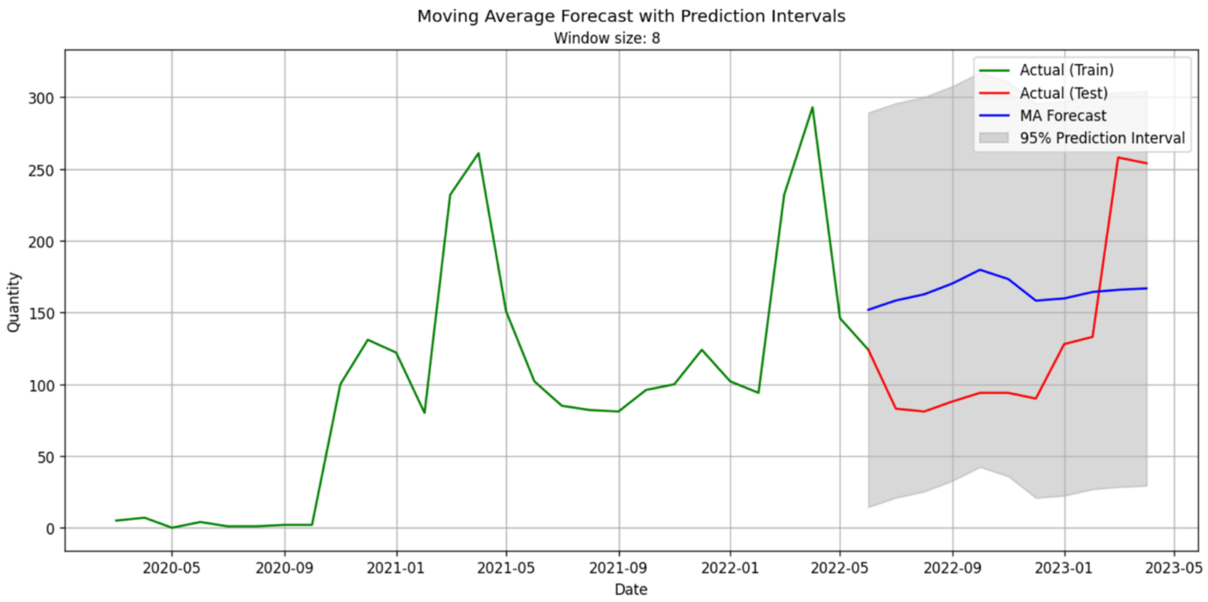
What is SMA?

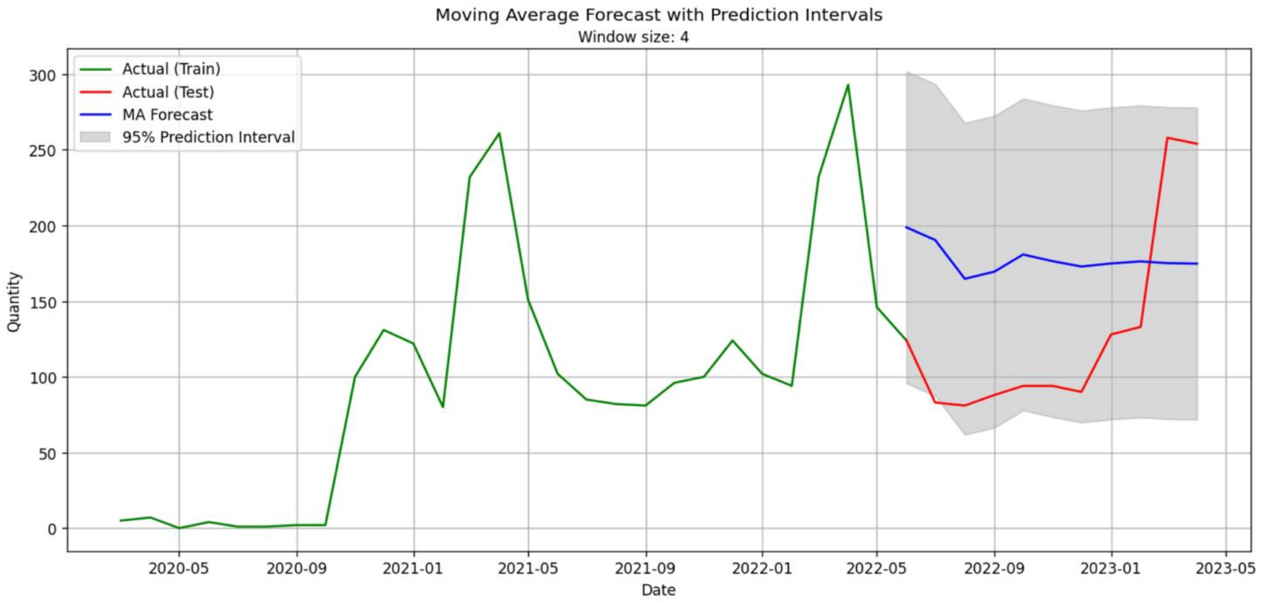


# *Starting with Simple Moving Average (SMA)*

And results…

- Improved version of MA model (95% prediction interval), Prediction Interval – 95% chance forecasted values within range.
- Assumption - Normality in data, Window size - 4 & 8 months



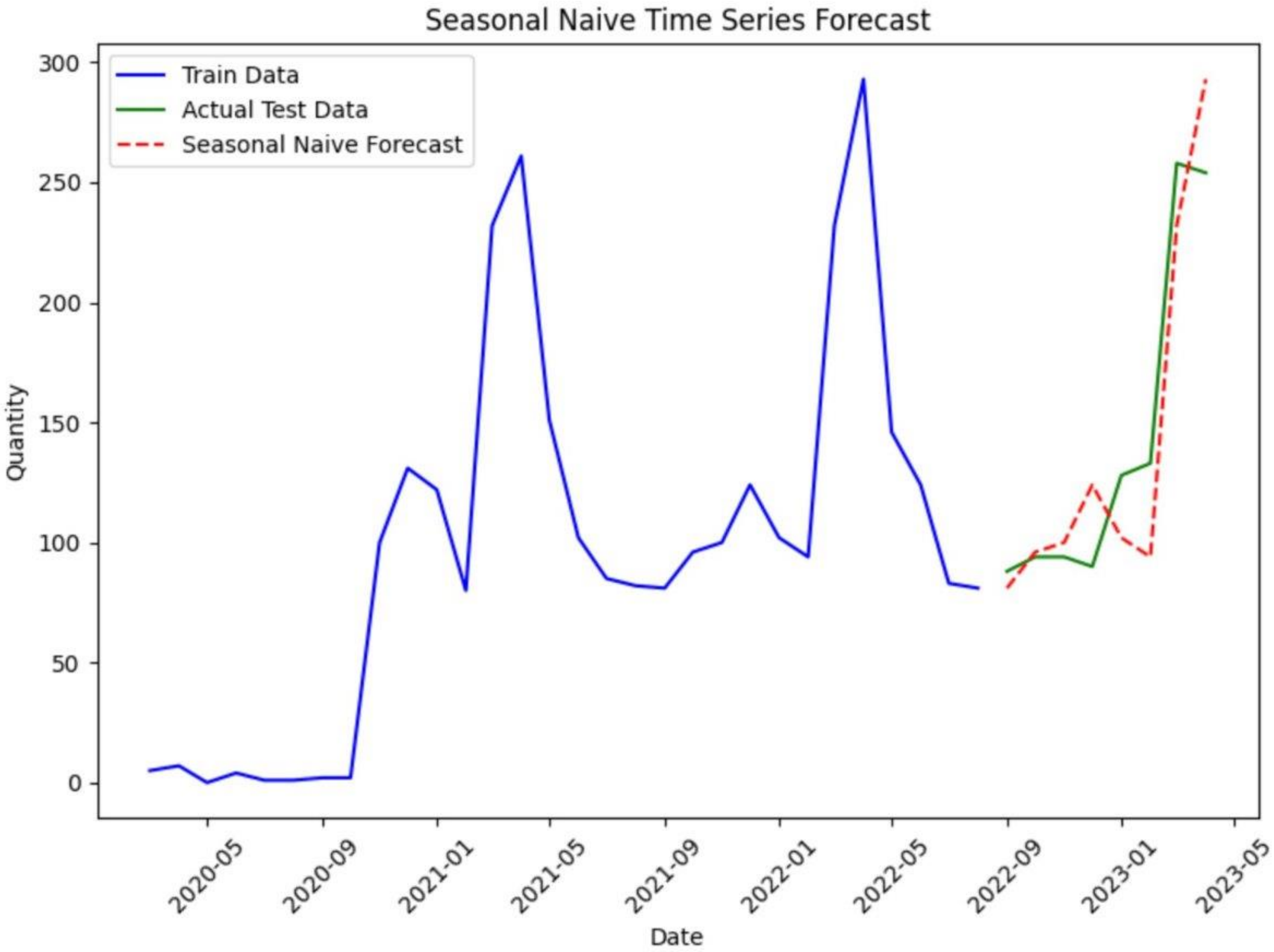



# *Moving onto Seasonal NAÏVE forecast*

And results…

- Assumption – current year's demand resembles last year's demand pattern
- Fails to consider changing trends
- Benchmark model similar to SMA
- Error Metrics:

| Metric | Error |
|---|---|
| RMSE | 26.55 |
| ME | 2.12 |
| MAE | 22.38 |
| MSE | 704.88 |

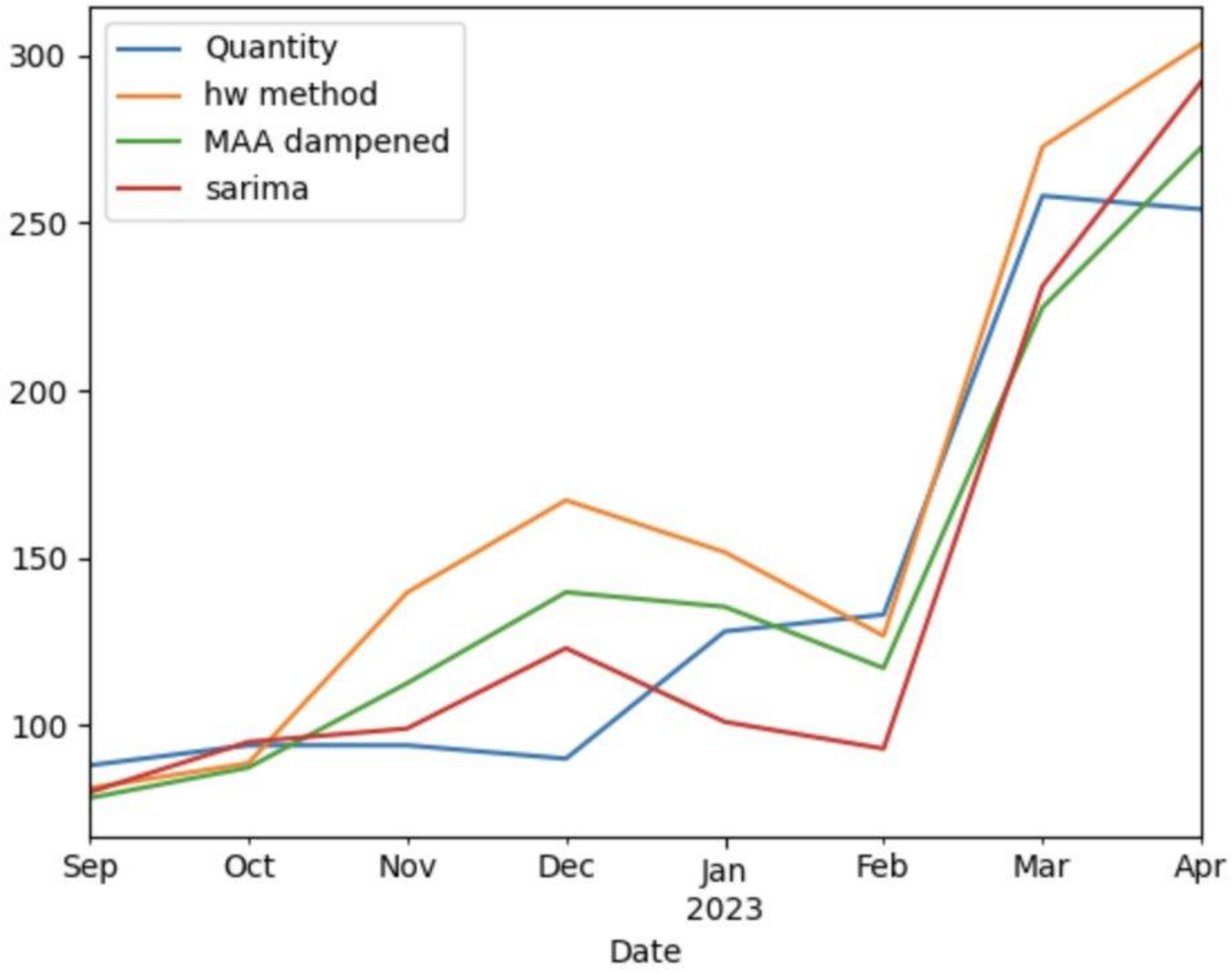


# *Moving onto Exponential Smoothing…*

What is Exponential Smoothing?

- Different Models Based On Error, Trend and Seasonality
- Utilizing Additive and Multiplicative Factors to build model
- Utilizing Dampened Trend to account for slowing trend over time

- Transformation on multiplicative Factors

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# *Exponential Smoothing forecast*

And results…

- Observation - AAA model captured seasonal peak for March
- Overprediction – Oct 2022 to Jan 2023
- Negatively correlated for Oct 2022 to Jan 2023
- Trend is captured accurately
- Seasonality correctly captured for Feb 2023 to Mar 2023
- Seasonality misses trough for Dec 2022

| Model * | RMSE | ME |
|---|---|---|
| ANN | 91.3 | -61.4 |
| AAN | 79.6 | -49.5 |
| AAA | 37.7 | 24.0 |
| AADA | 27.1 | 3.3 |
| **MADA** | **24.3** | **3.5** |

* ANN – level; AAN – level, trend; AAA – level, trend, seasonality, AADA – damped trend, MADA – multiplicative and damped trend

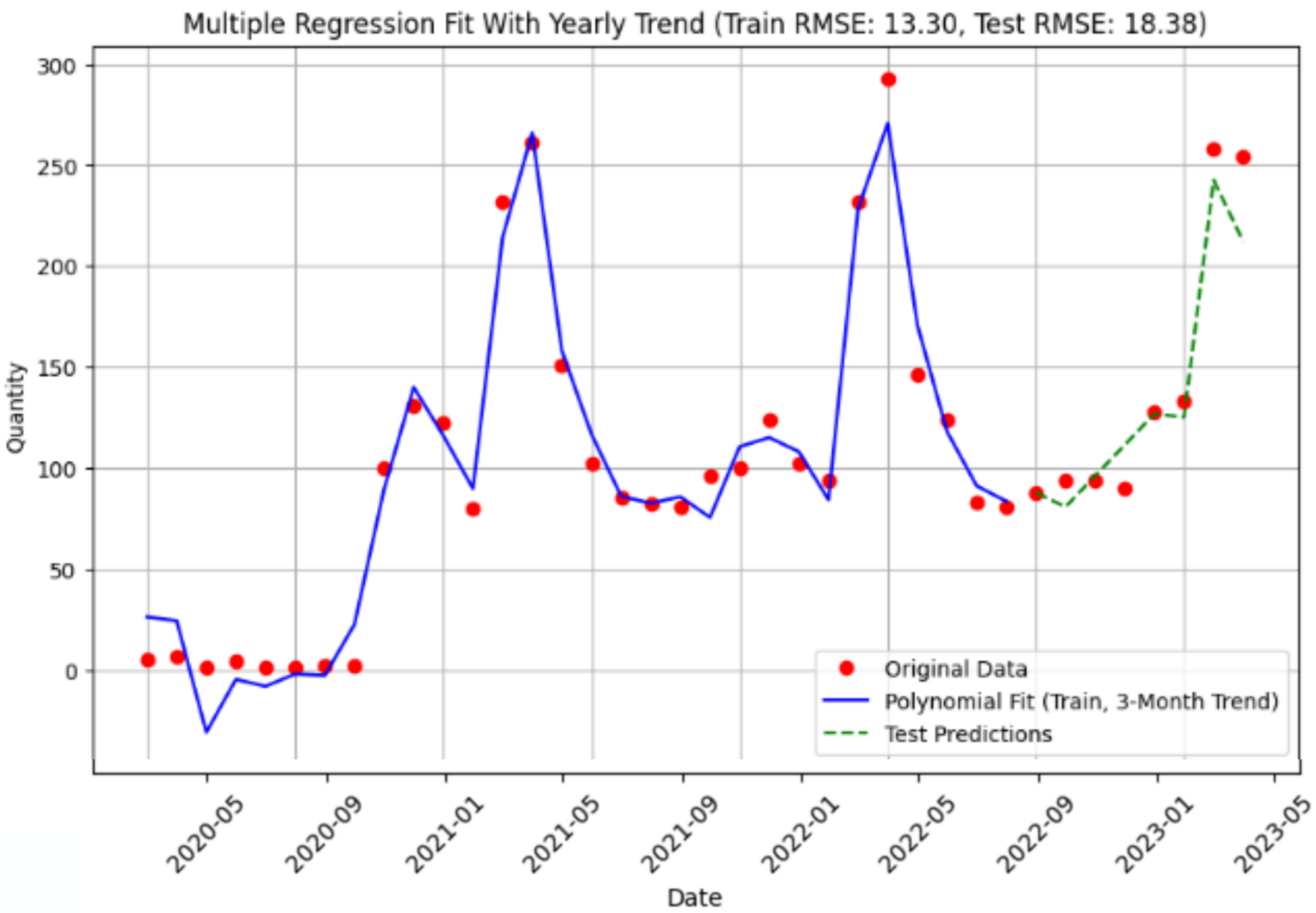
# *Moving onto Regression...*

What is Regression?

- **Polynomial equation**
  - Coefficients are calculated based on smallest sum of squared residuals (least error)

Error Metrics:

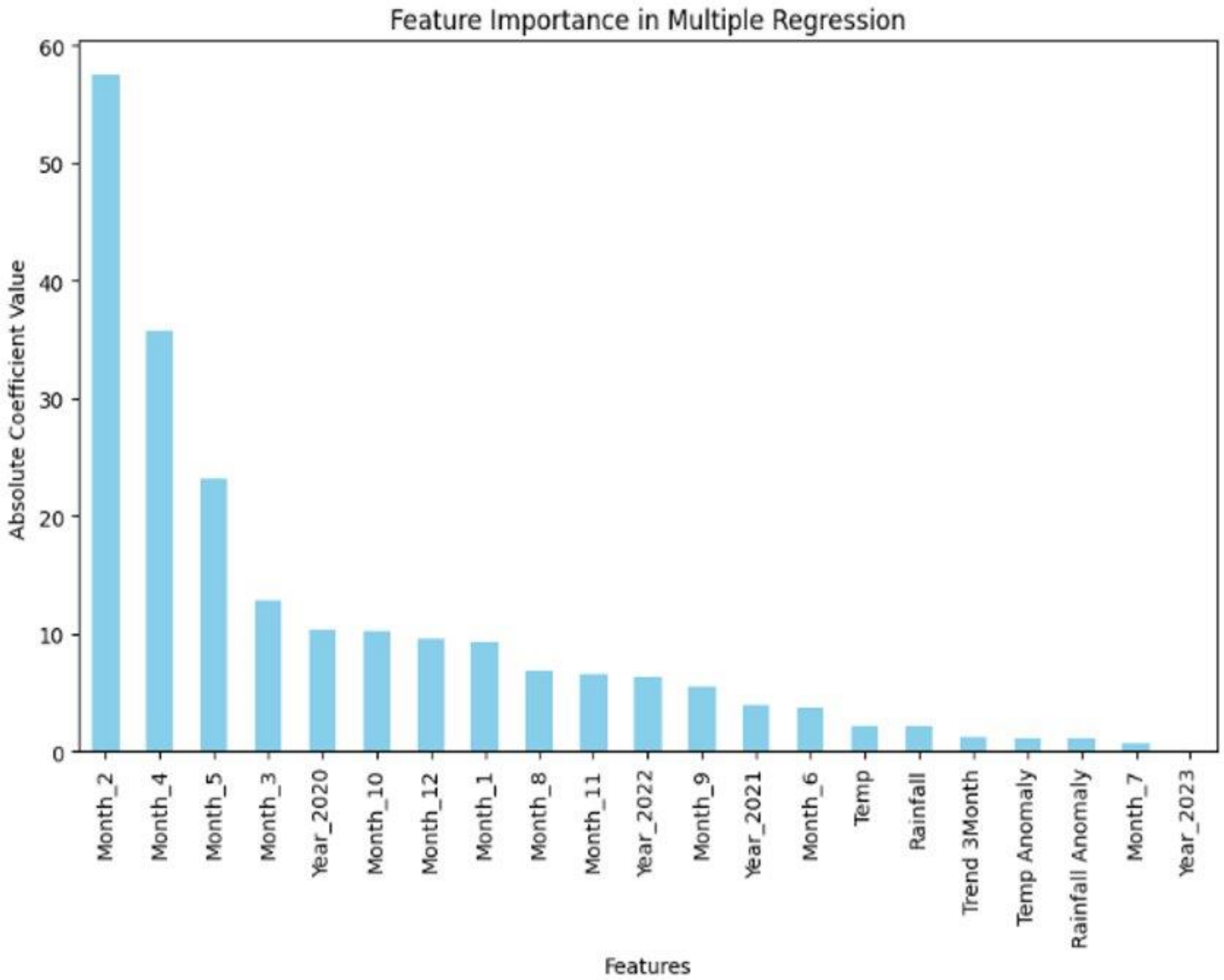
- RMSE: 18.38
- ME : -7.05



# *Regression forecast*

And results...

- Feature importance is determined by the magnitude of their coefficients
- Federal Funds Rate dropped, because it significantly reduced model accuracy
- Beyond Month and Year, **Rainfall** and **Temperature** are significant features



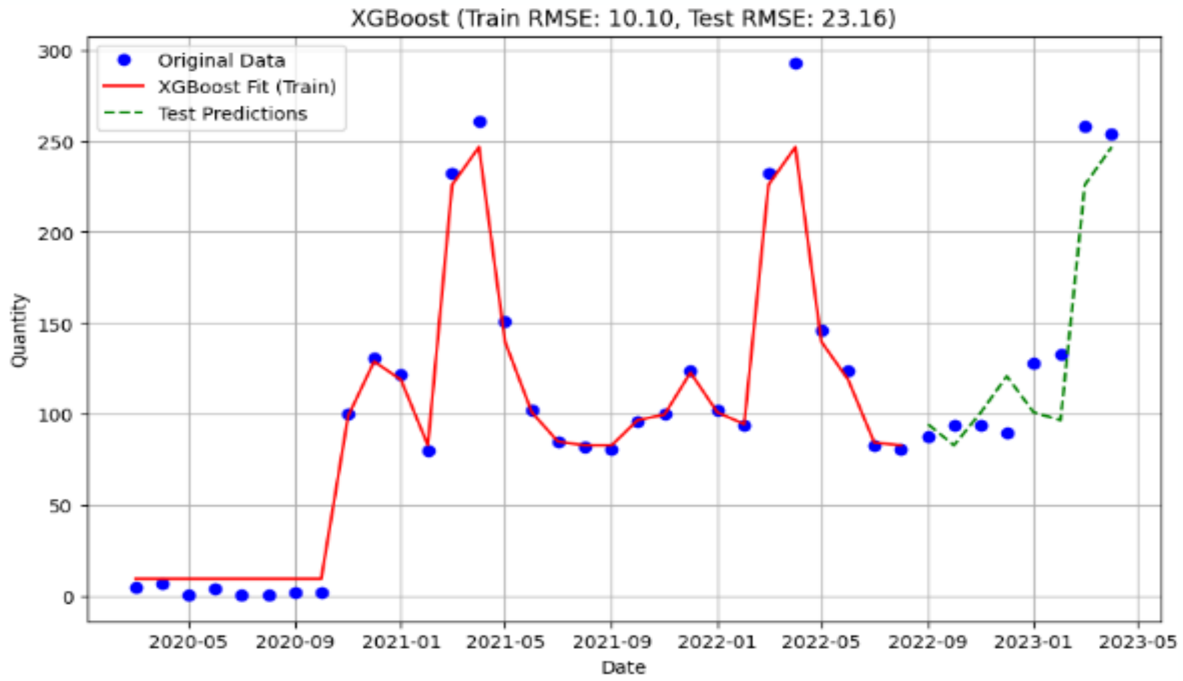
# *Moving onto XGBoost...*

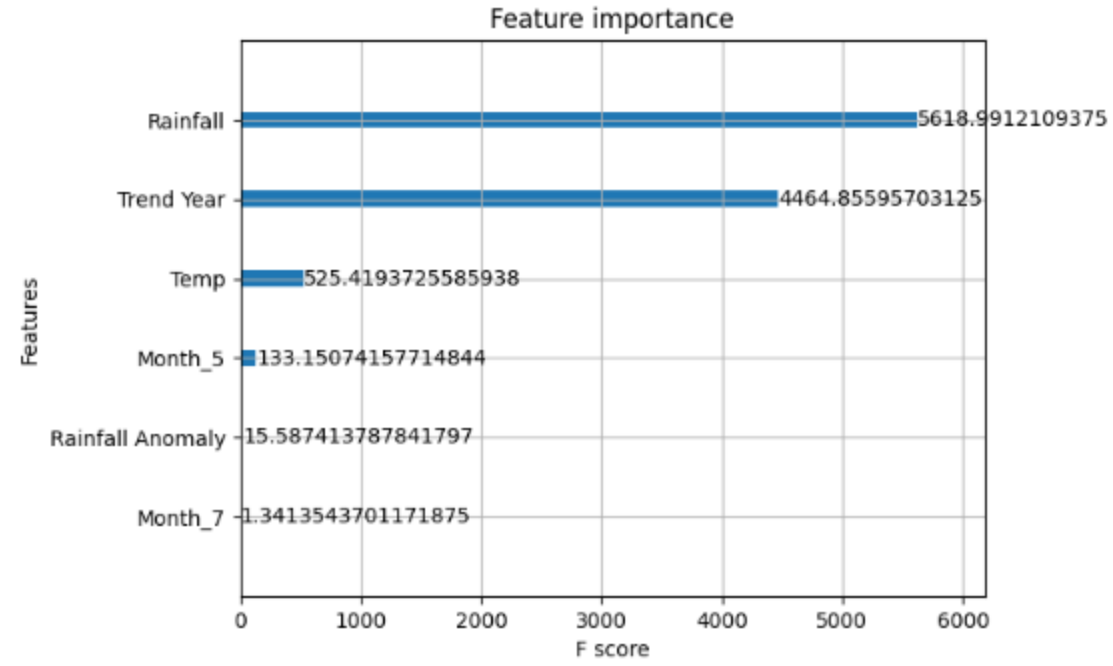
What is XGBoost?

- XGBoost works by making predictions and then calculating the residuals.
- It will then use a decision tree to reduce the error.
- It will then keep iterating through this process until it has completed a certain number of trees.

# XGBoost Forecast

And results…



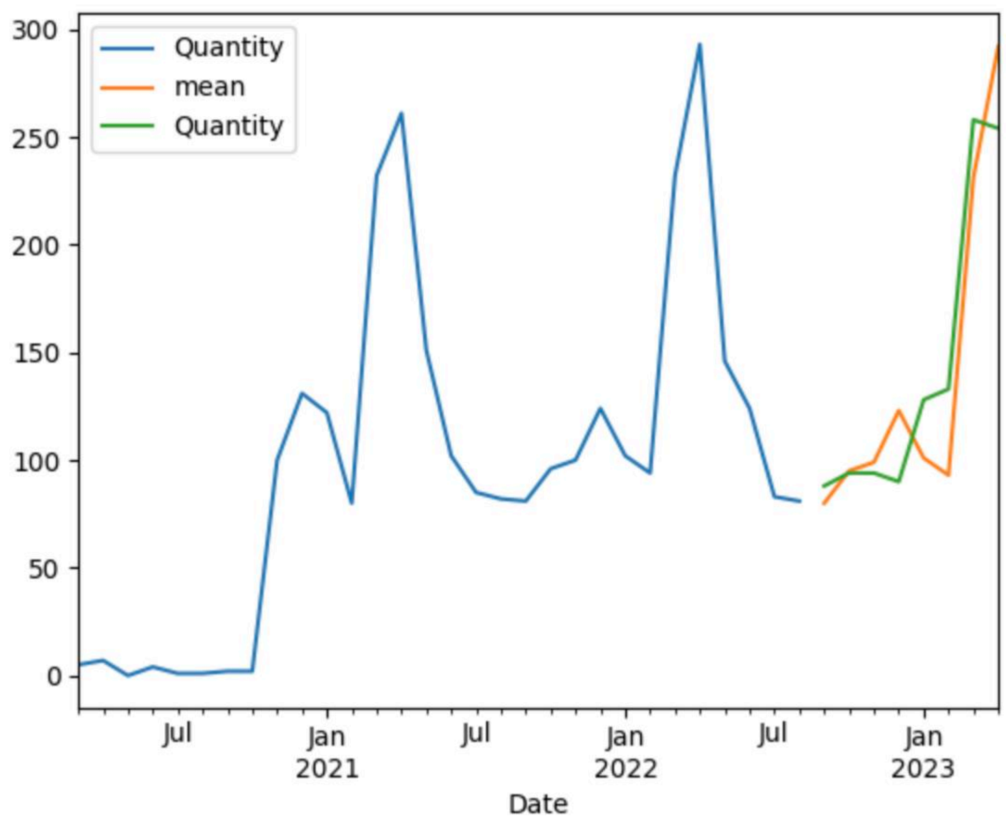




- Error Metrics: RMSE = 23.16, ME = -8.65
- The negative mean value means that this model is likely to underpredict the amount of demand.
- Rainfall and yearly trend are highly used for making predictions

# *Moving onto SARIMA/SARIMAX...*

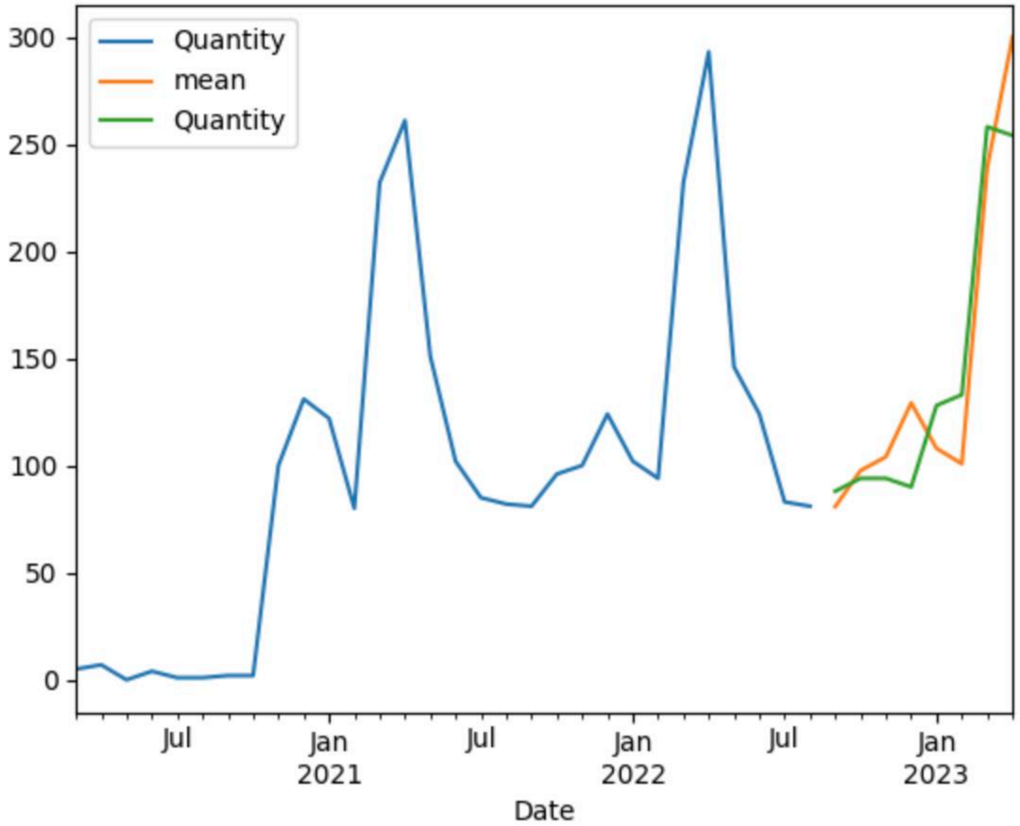
What is SARIMA/SARIMAX..?

**(No External Features)**




**SARIMA(0,1,0) (0,1,0,12)**

**(Federal Funds External Features)**




**SARIMA(0,1,0) (0,1,0,12) + Federal Funds**

Addition of Federal Funds as a feature improved model accuracy and helped eliminate underprediction

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# *SARIMA/SARIMAX forecast*

And results...

- External predictors into the SARIMA model, like Rainfall and Temperature, increased the RMSE of the model and produced a negative ME. Similar effect was observed with the introduction of Rainfall Anomaly and Temperature anomaly

- The only predictor that had a positive impact on the performance of the model was the inclusion of the federal funds rate. Improving the RMSE and producing a ME that was indicative of more overprediction from the model

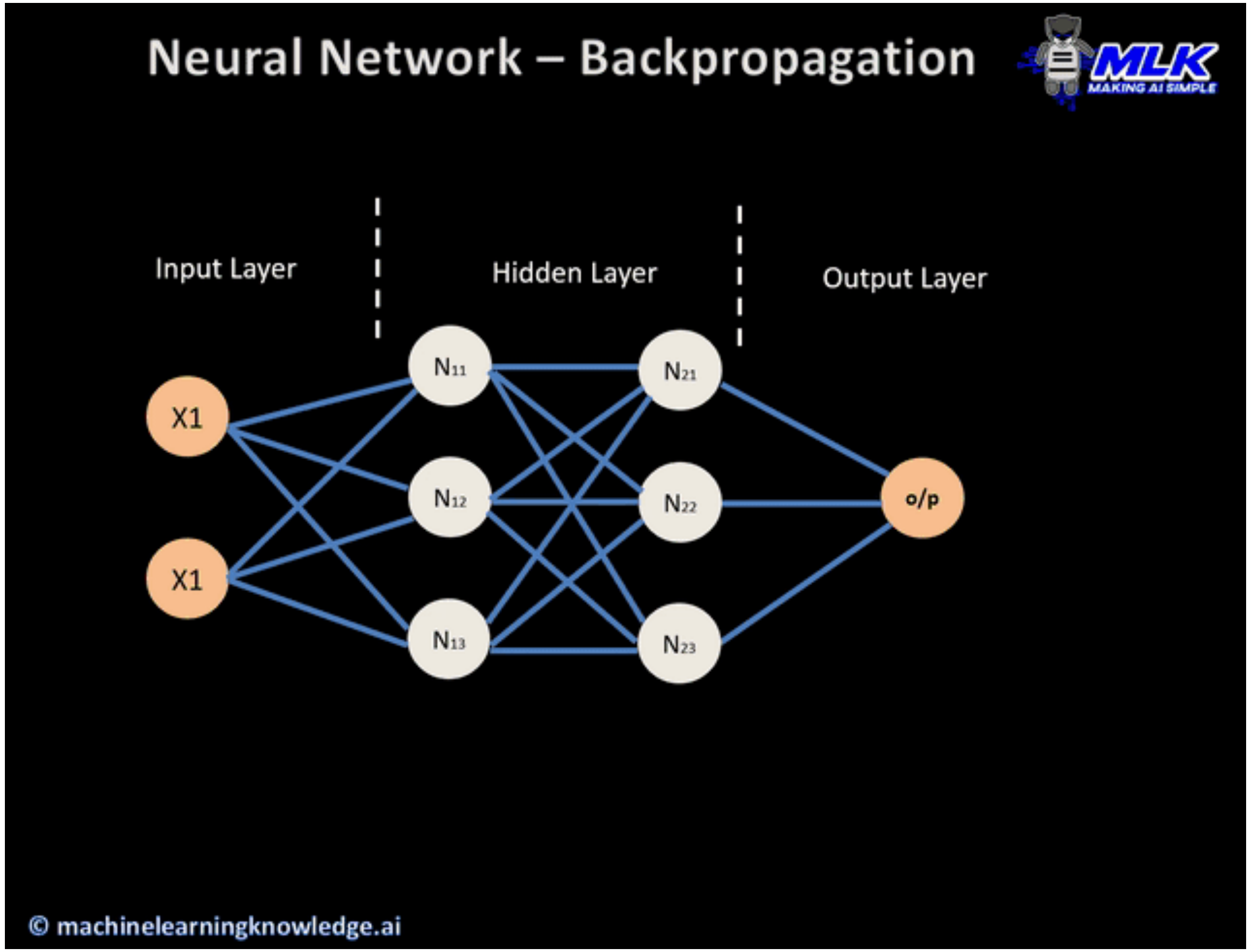
| Model | RMSE | ME |
|---|---|---|
| RTF | 35.827 | -15.756 |
| RF | 28.623 | -5.045 |
| R | 28.744 | -5.722 |
| SARIMA | 26.648 | -3.125 |
| F | 26.490 | 2.514 |

- RTF – Rainfall, Temperature, Federal Funds
- RF – Rainfall, Temperature
- R – Rainfall
- SARIMA – No External Features Used
- F – Federal Funds

# *Moving onto Neural Networks...*

## What are Neural Networks?

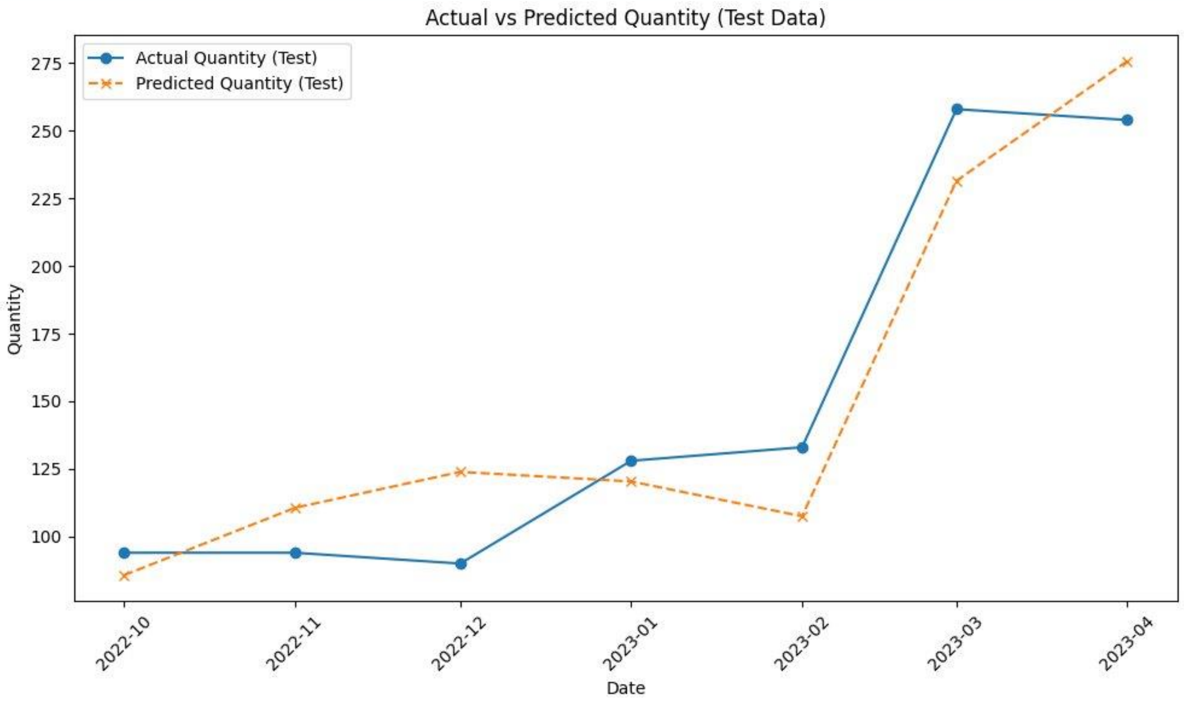
- Input Layer (Features),
- Hidden Layer (Analysis & Processing), Output Layer (Predicted Demand Quantity)
- Every input is from the output of a prior layer
- Activation Function: function that calculates the output of the node based on its individual inputs and their weights
- The type of Neural Network used is called a Dense Layer Neural Network
- **Hyperparameters:** Activation Function: ReLu, Learning Rate: 0.001, 300 Epochs , 4 Hidden layers, with 32, 64 , 128 , and 64 nodes, respectfully.
- **Challenge:** Difficult to gauge robustness of model due to lack of data

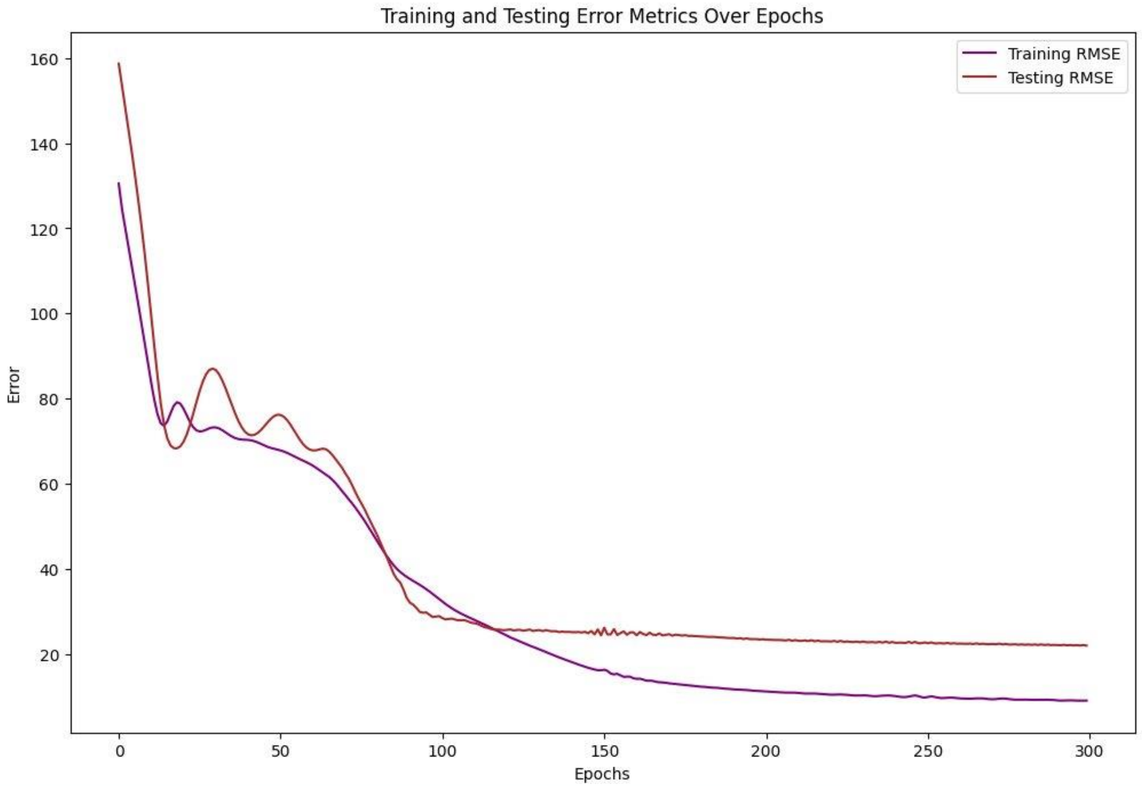


# *Neural Networks forecast*

And results...

- **RMSE:** `21.9795`
- **ME:** `0.6184`

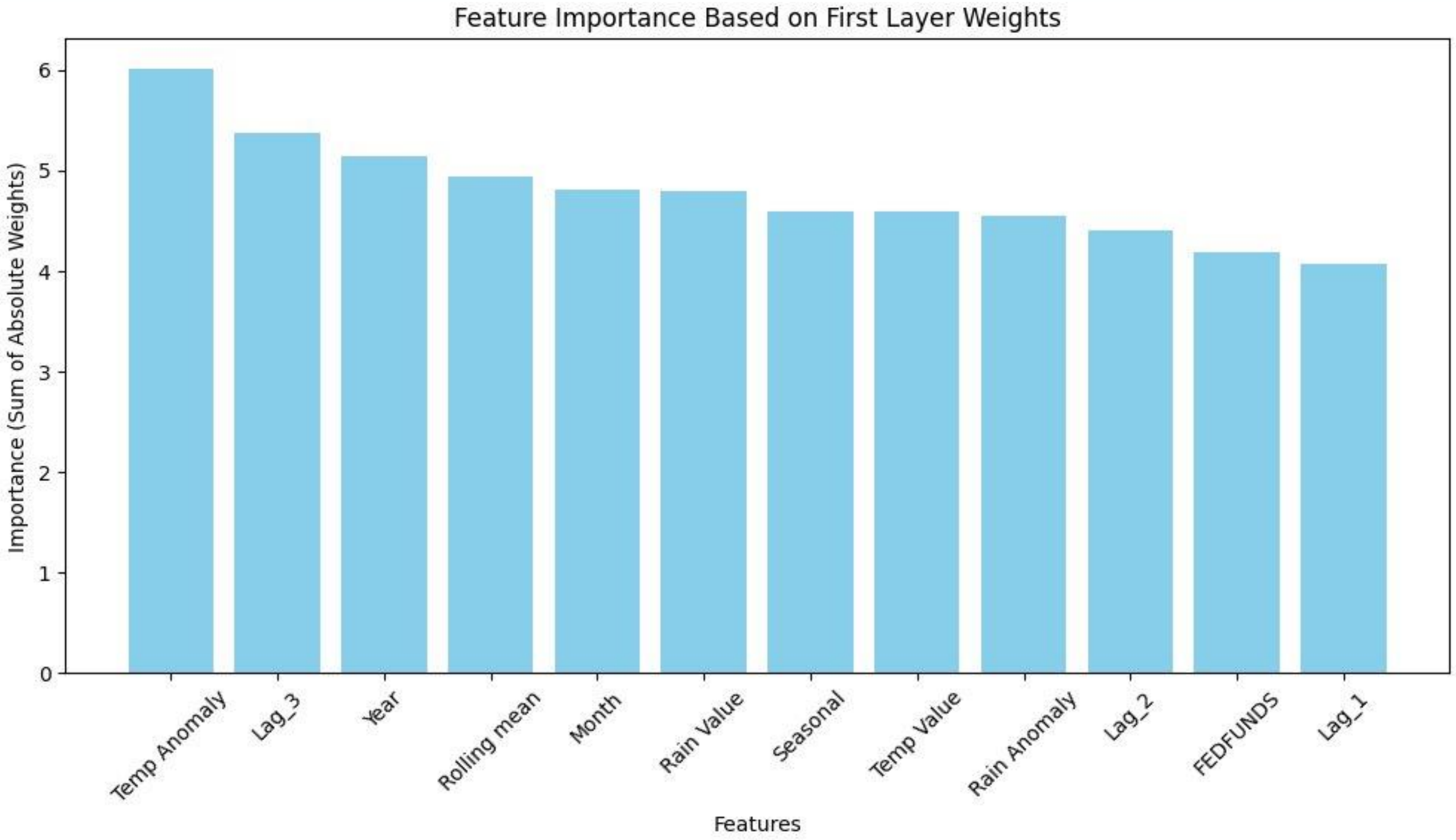


# *Neural Networks forecast*

And results…

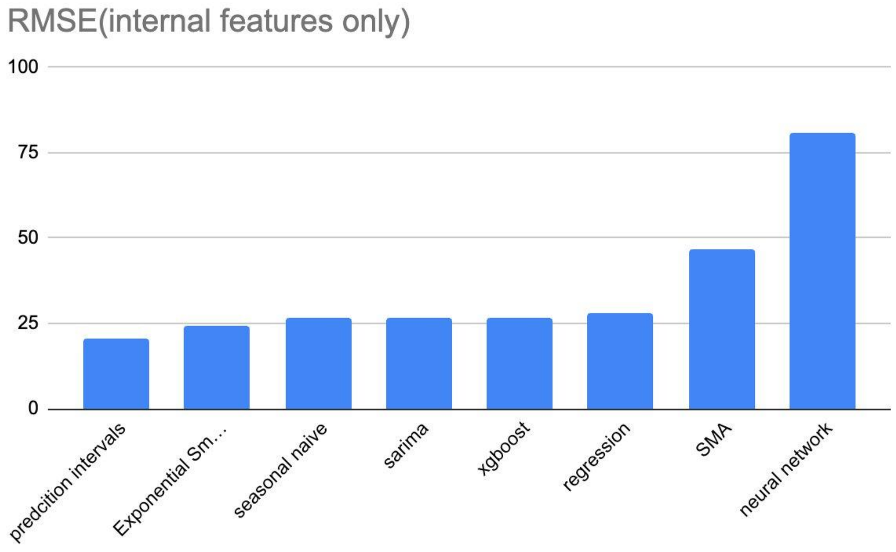
- **Key Features Identified:** Temp anomaly, year , 2 month rolling mean
- **Model Limitations:** Limited data impacted accuracy of feature importance rankings
- **Caution in Interpretation:** Feature importance results might not fully reflect true impact due to data constraints

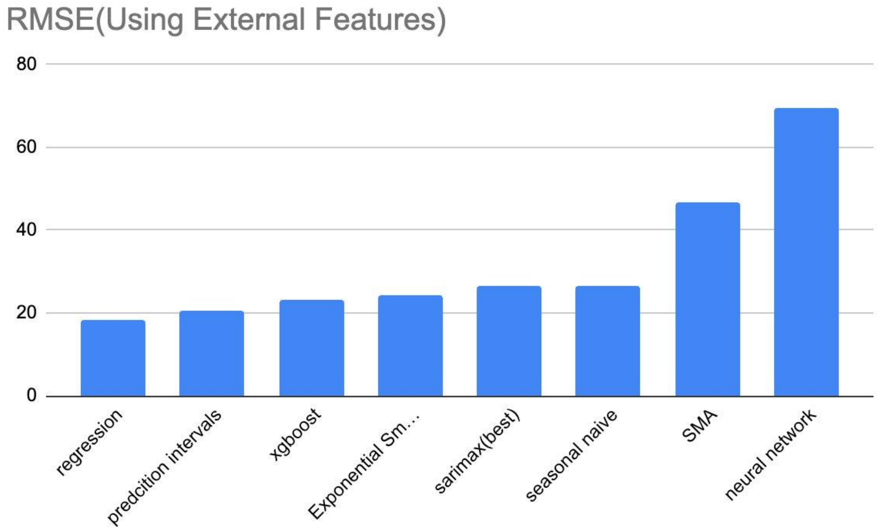


# *Time for Model Comparisions!*

Which ones preform the best?

- Almost all models outperformed benchmark simple moving average model
- Exponential Smoothing and XGBoost showed **strong performance against benchmark model** and serve as strong foundation for further model development
- Testing on additional time series can further validate model results

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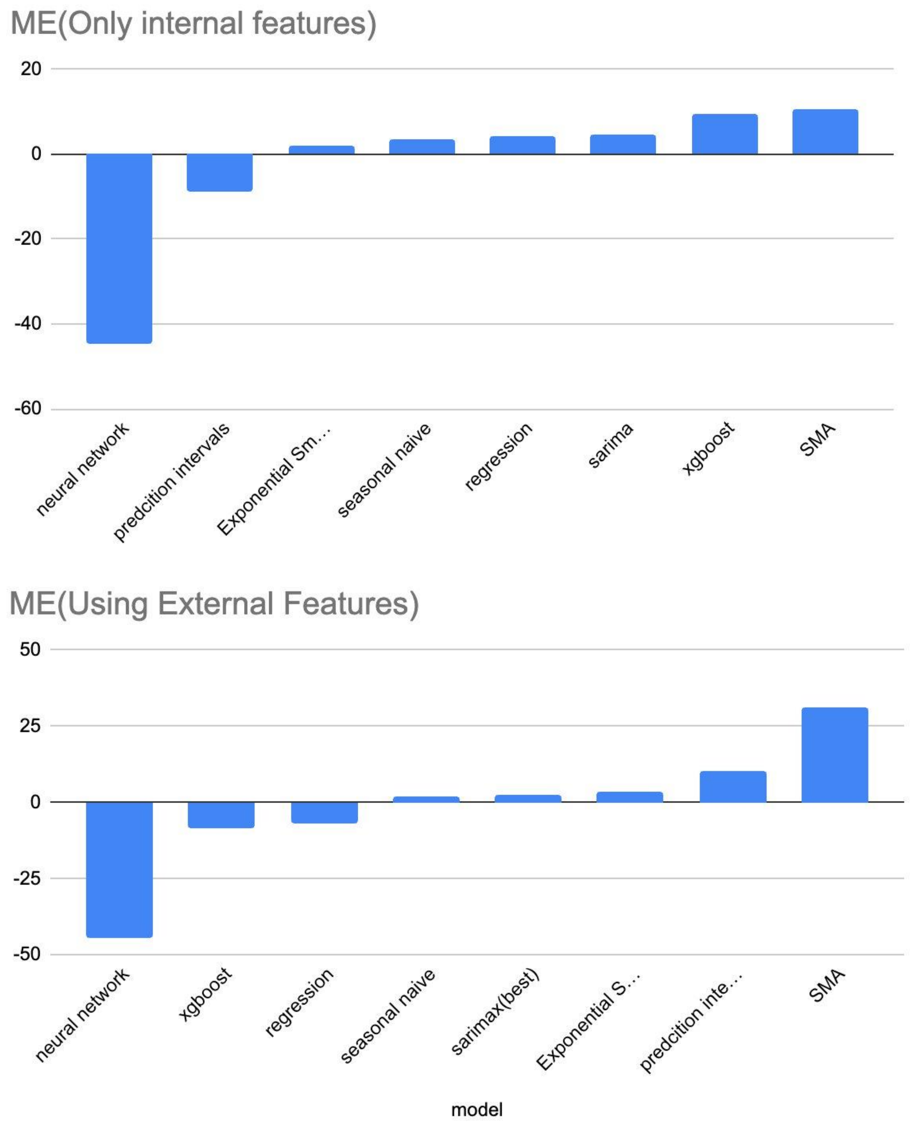
# *Time for Model Comparisions!*

Which ones preform the best?

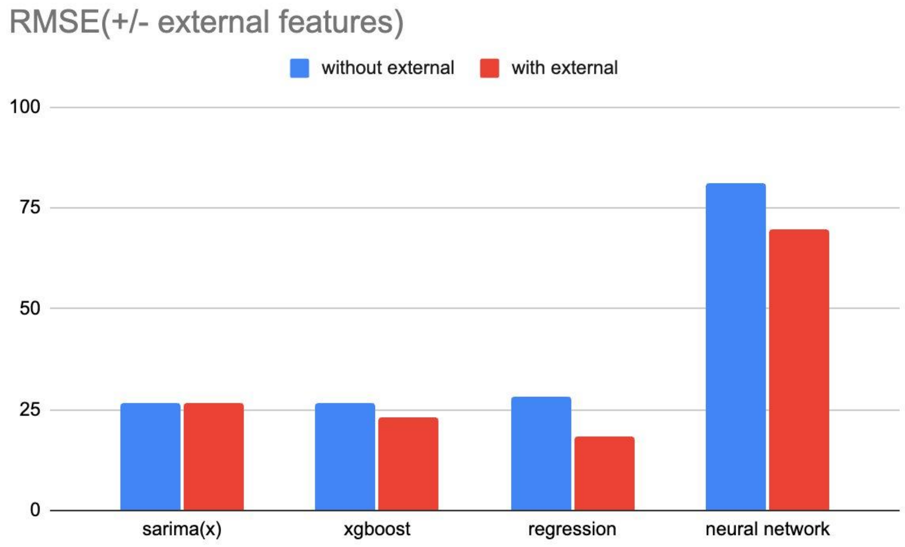
- Almost all models without external features showed consistent **overprediction** as compared to benchmark models
- Systemic **underprediction** can cause locations to undersupply and fail to capture real consumer demand. Results in lost revenue and produces poor data for future use.
- Consistent overprediction shows the need of further model development.

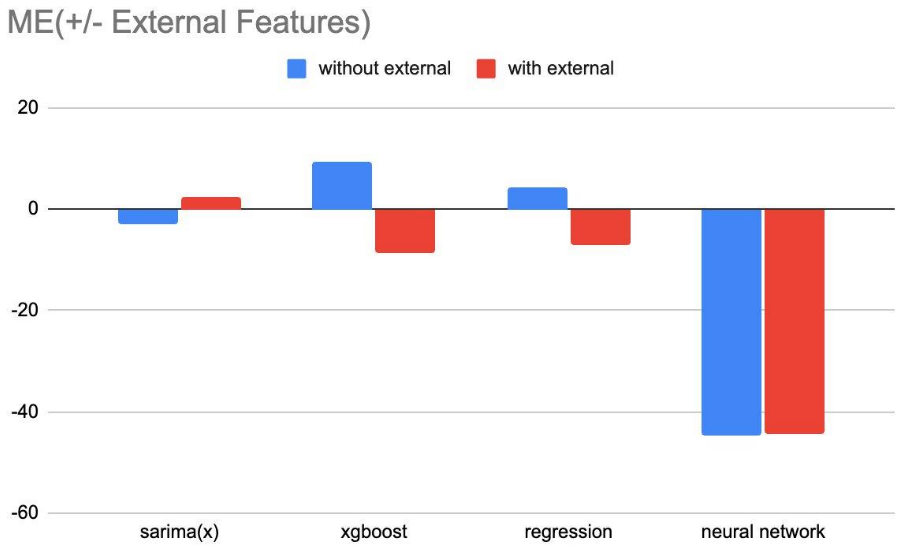


# External Features Impact on Model Performance

How do they change things?

- SARIMA, XGBoost, Regression, & Neural Networks which were the models that could incorporate external features clearly showed a decrease in RMSE
- On the other hand there was also a greater propensity for underprediction following such improvements in RMSE reflected in the ME
- This could suggest that there does need to be further development of external features, as while RMSE was increased, underprediction can be more costly than less correct overpredictions



# *Results and Conclusion*

## So what?

- Multiple regression (MLR) - least RMSE with a value of 18.63.
- Exponential smoothing model had the best ME with a value of 3.529
- Neural Network : the highest ME value at ~50 and highest RMSE value at 69.5280. This is due to limited data for training
- Conclusion: Although MLR performed the best, generalizing would require testing on multiple part location combinations

Point Forecasting Model Exploration → Point Forecasting Model Generalization → Probabilistic Forecast Model Implementation

# *Limitations and Future Scope*

**What's next?**

Limitations

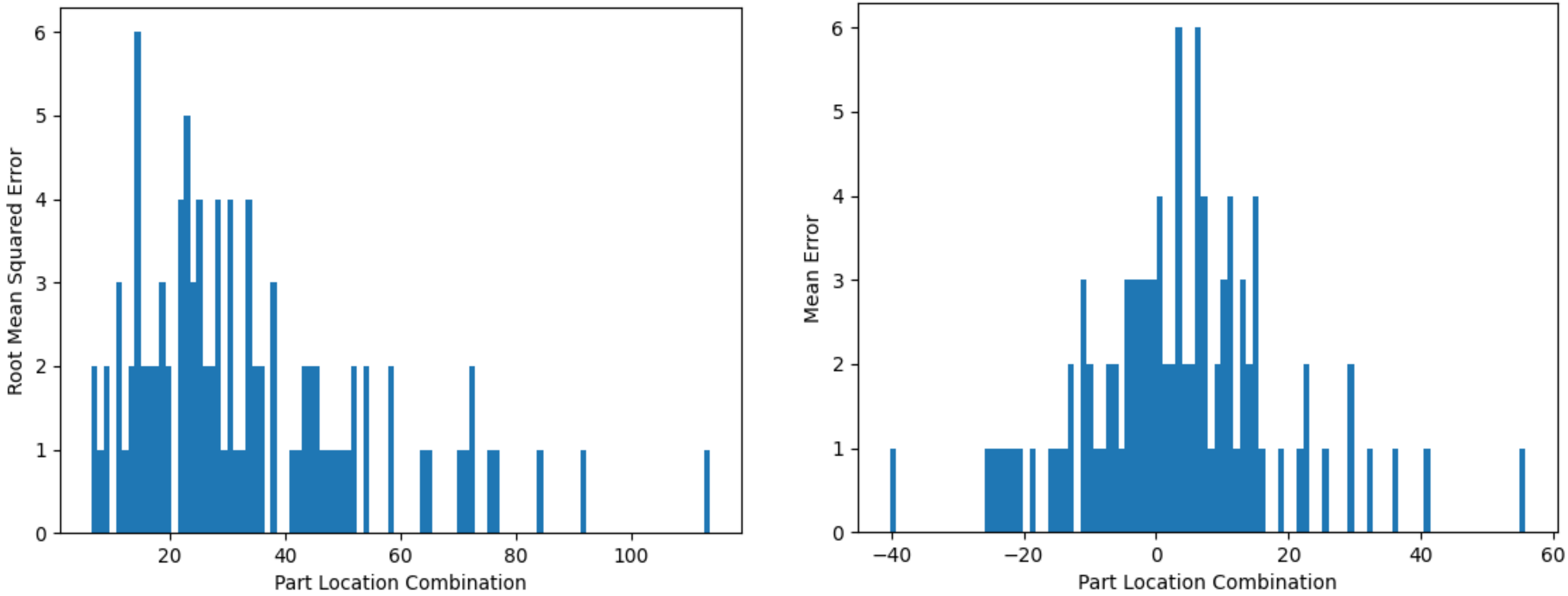
- Trained only on a single part location combination and only three years' worth of data
- Feature are time dependent, should have tested by lagging certain features for prediction

Future Scope

- Extend it to 100 part – location combination
- Probabilistic modelling – provides a range of expected demand values
- Quantile Regression, Bayesian Structural Time Series

# *Evaluation on 100 Time Series: SMA*

## Moving towards a more realistic scenario

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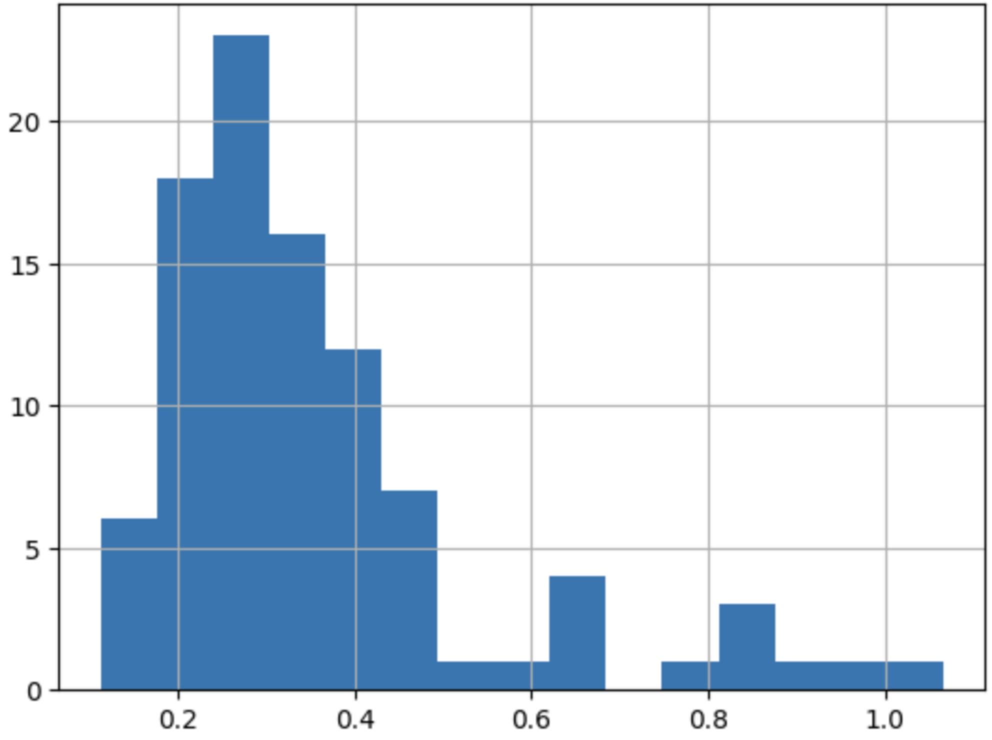
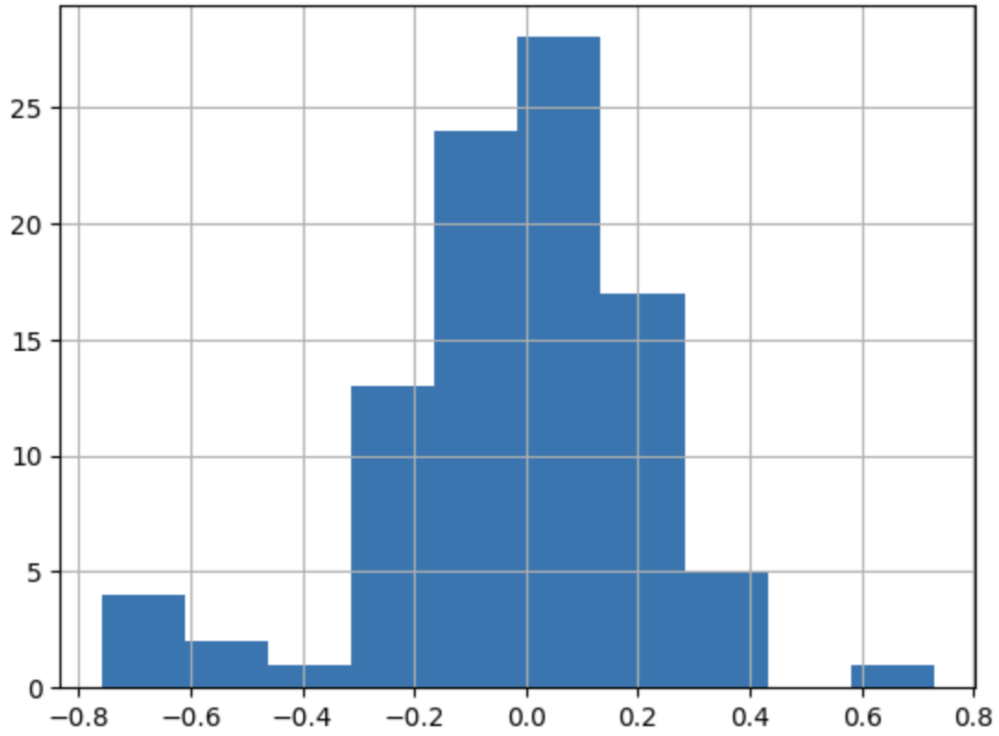
# Evaluation on 100 Time Series: Exponential Smoothing

## Moving towards a more realistic scenario

AA(dampened)A Model




RMSE
Avg: .35, Std: .19

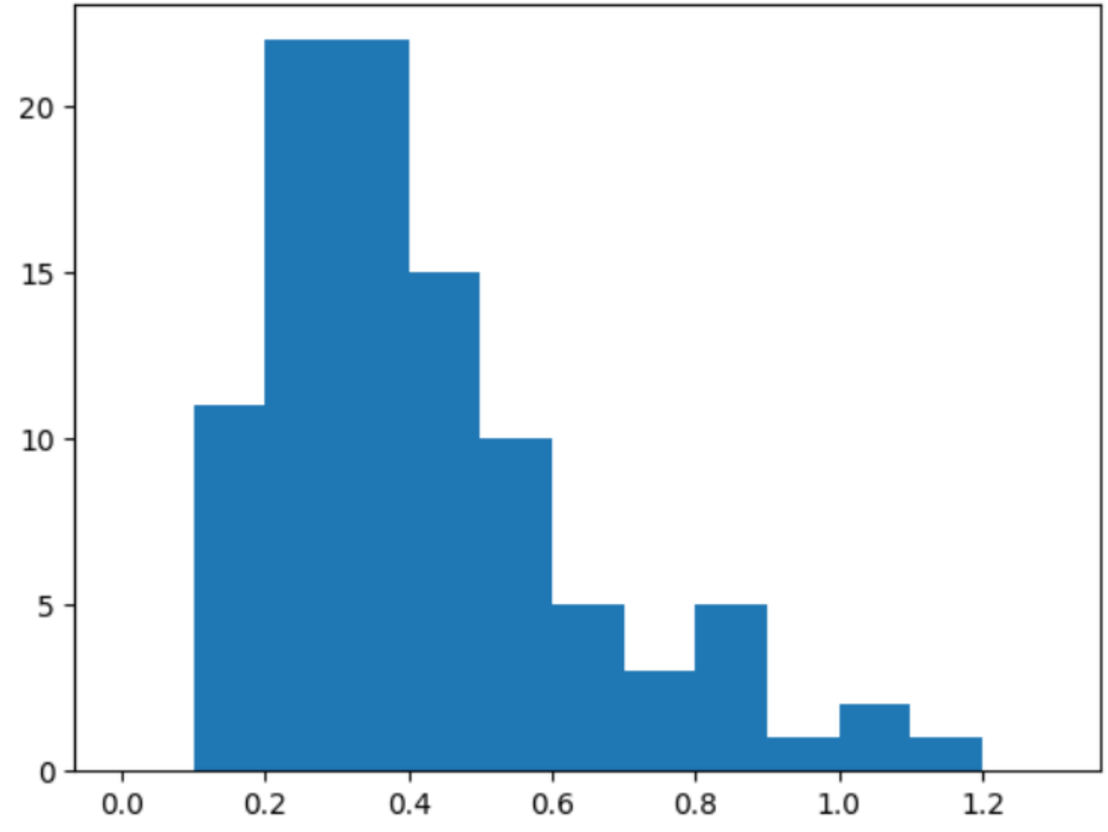



ME
Avg: -.016, Std: .239

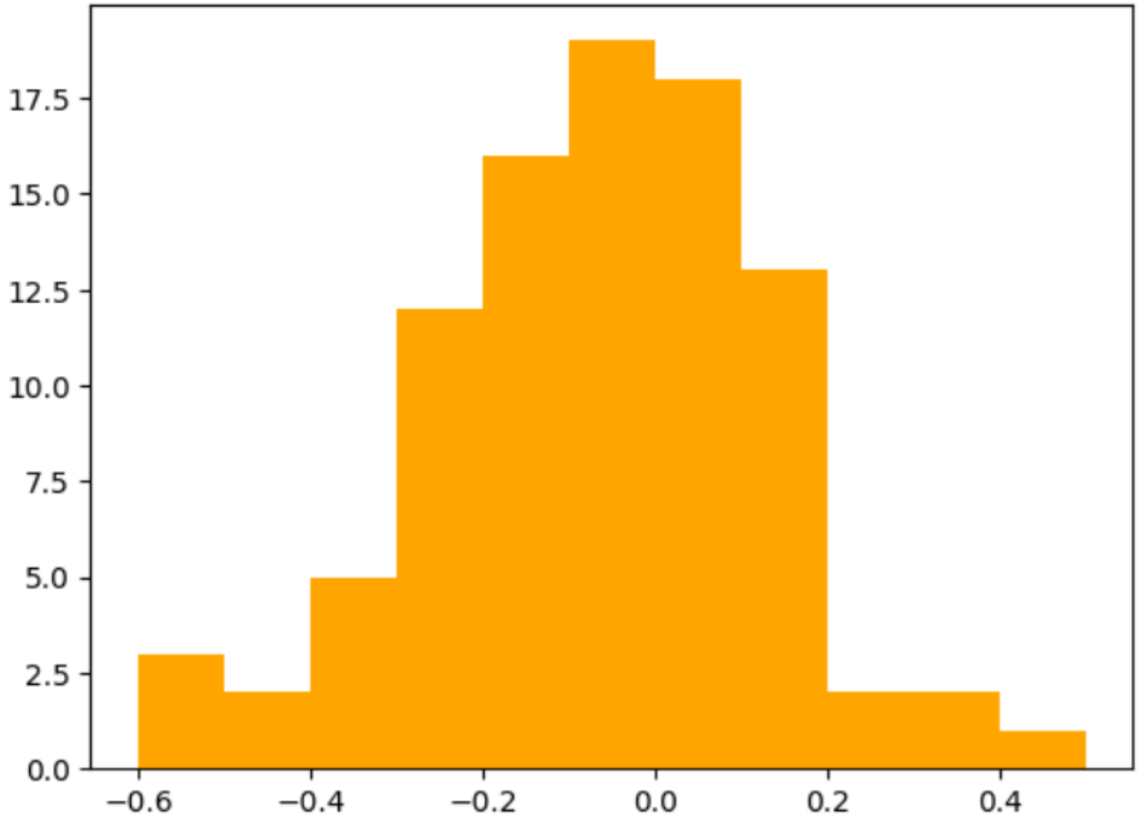
# *Evaluation on 100 Time Series: Regression*

## Moving towards a more realistic scenario

Normalized RMSE: RMSE / Mean(Demand Quantity)

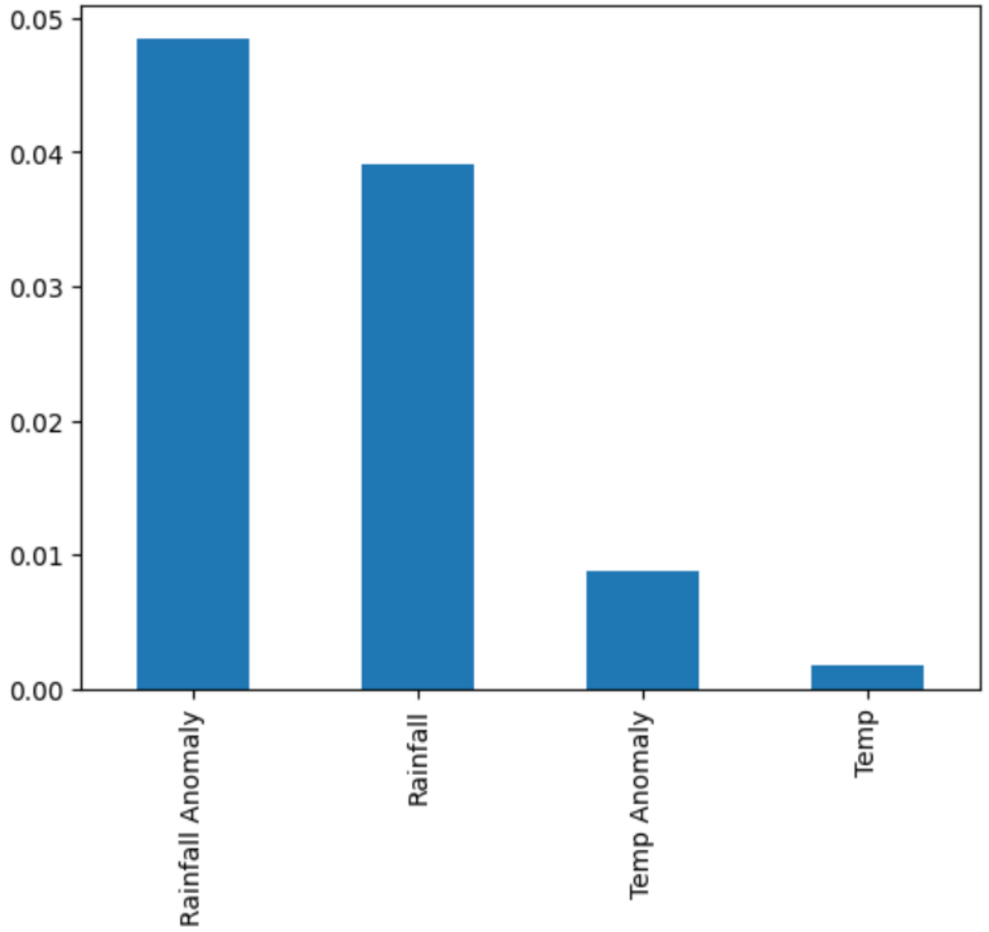



Relative ME: ME / Mean(Demand Quantity)



# *External Feature Importance: Regression*

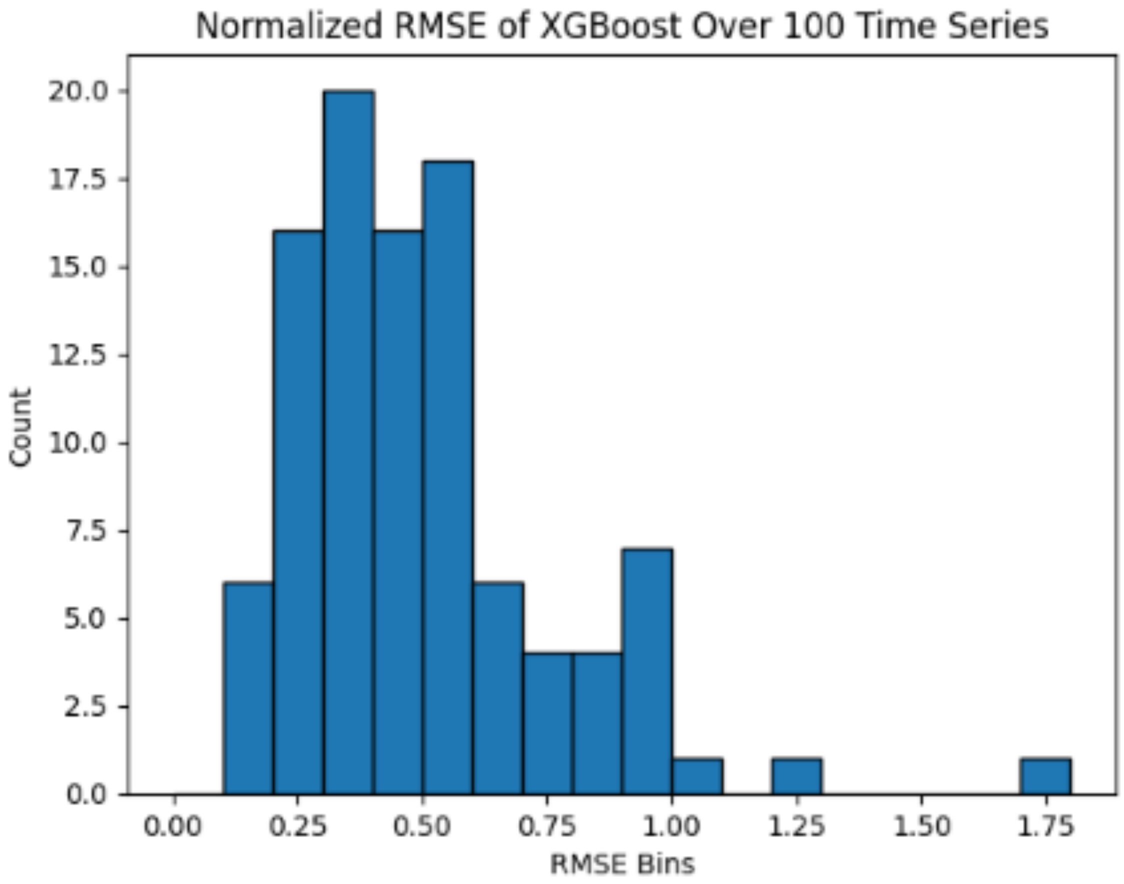
## Which Features are Essential to accurate predictions?

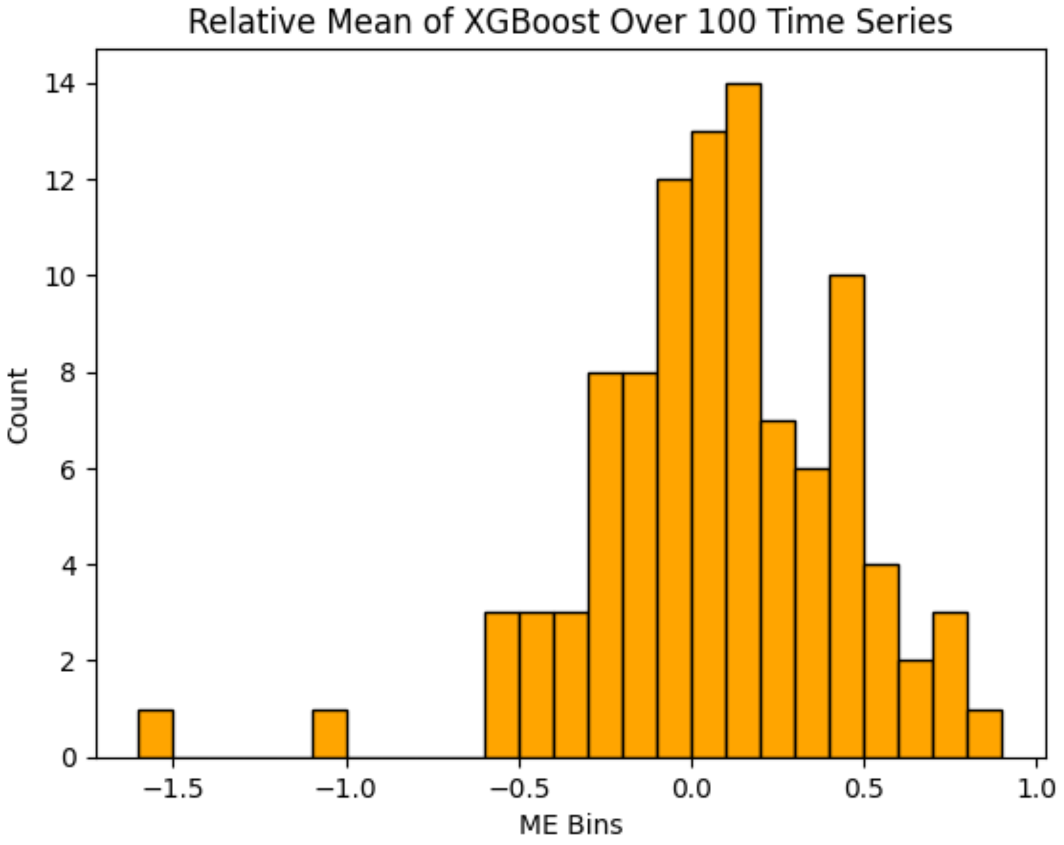



- Feature "Importance" percentages derived from magnitude of the polynomial coefficients
  - Totaled and divided across all 100 time series
- Trend and Seasonality dropped due to uncertainty on how to provide it as input
  - Model accuracy very minimally affected, error distribution slightly improved
- Months and Years excluded (mandatory inputs)
- Rainfall Anomaly (deviations of rainfall from long-run averages) and Rainfall have 4-5% importance in the prediction
- Temperature less important (< 1%)

# *Evaluation on 100 Time Series: XGBoost*

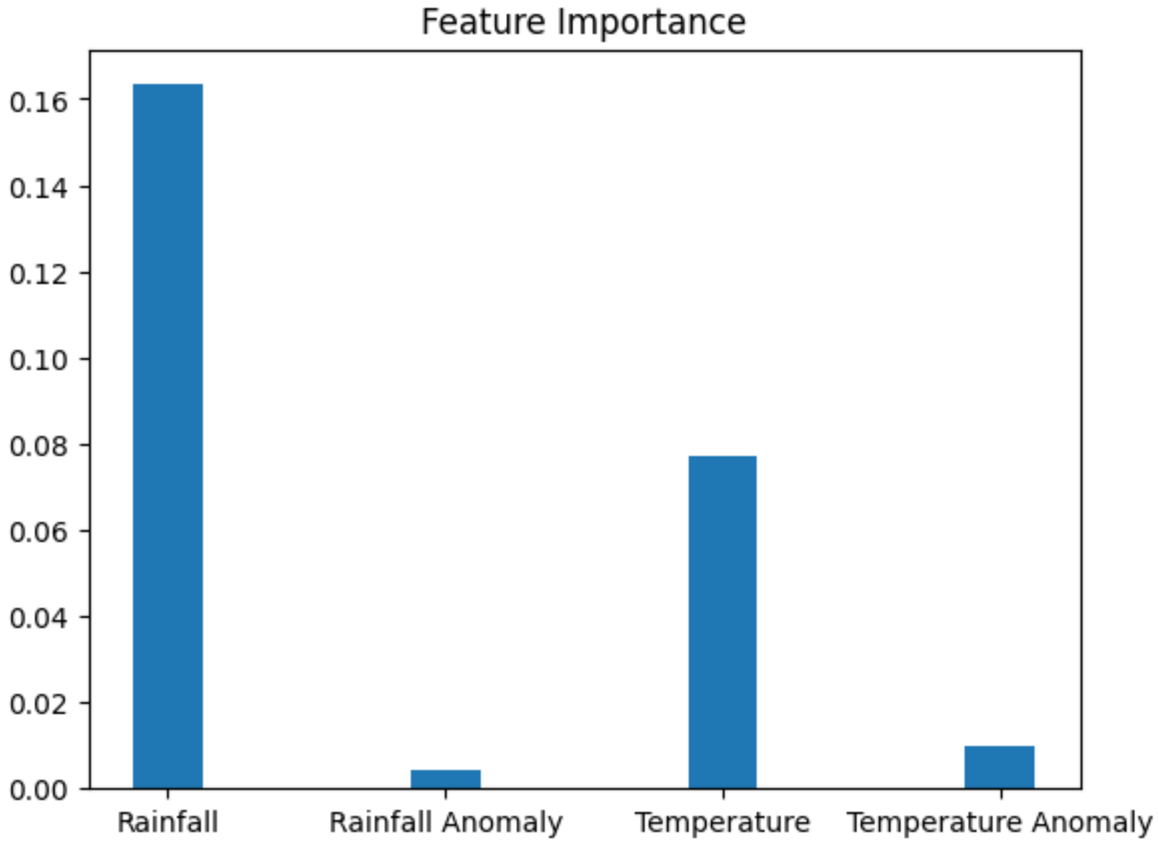
**Moving towards a more realistic scenario**



# *External Feature Importance: XGBoost*

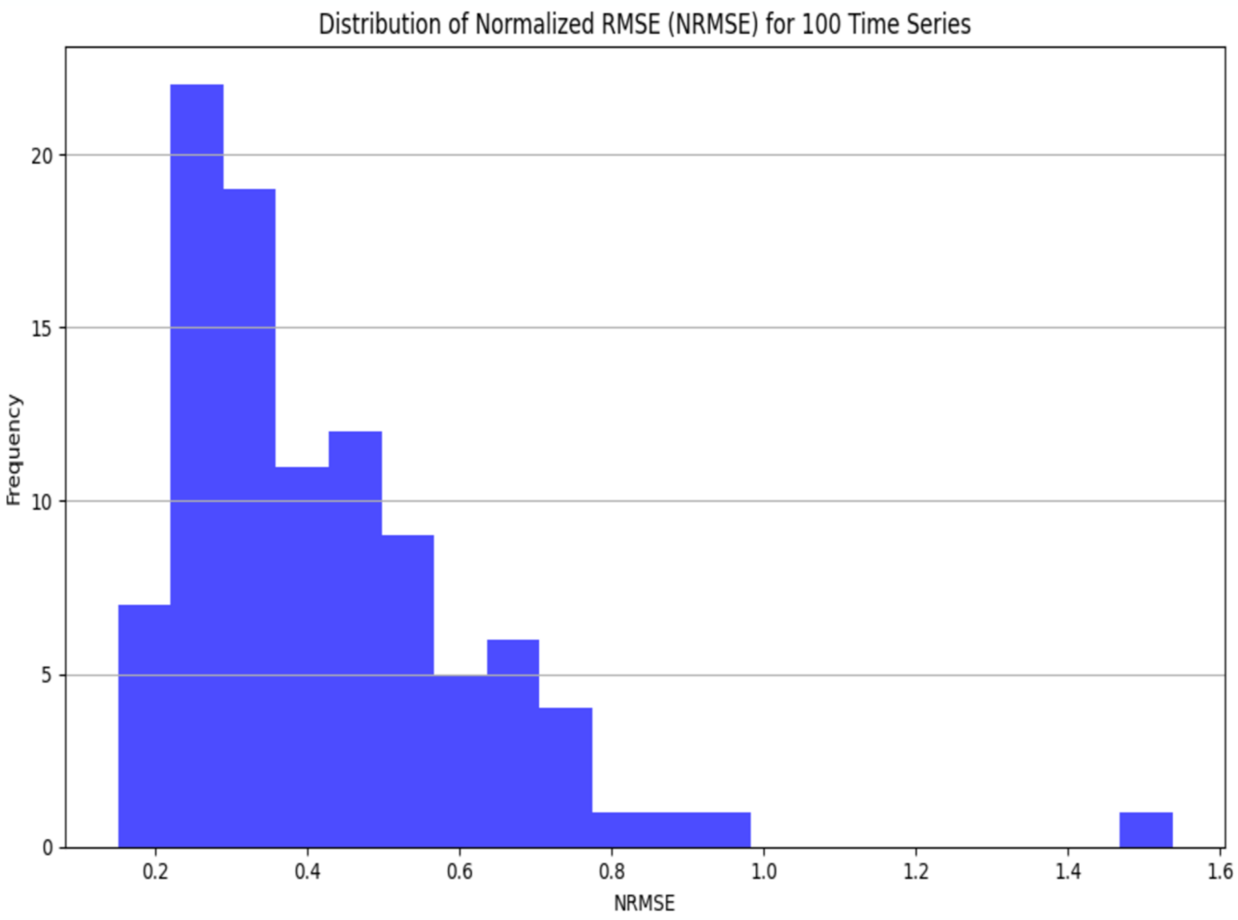
## Moving towards a more realistic scenario

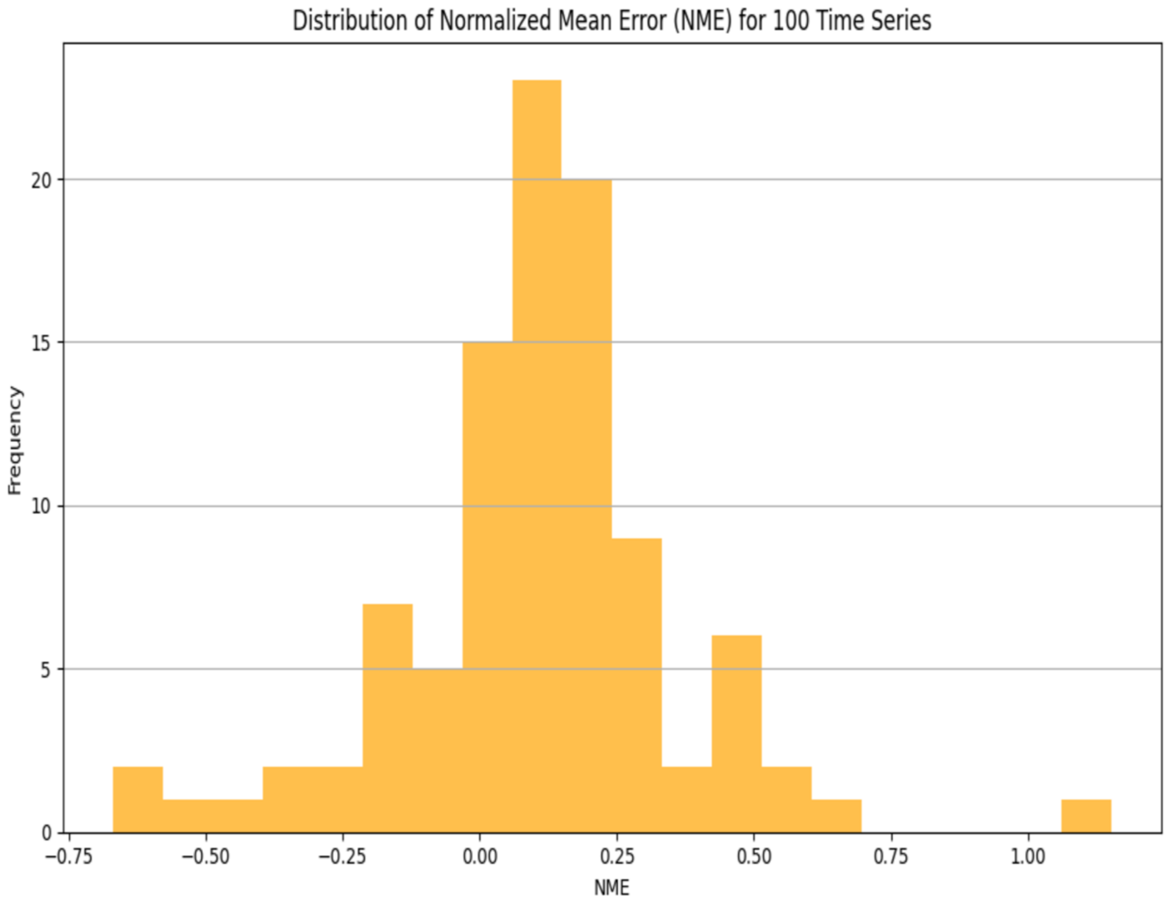



- Feature Importance was calculated by the number of times a feature is used to split data across all trees. Each importance was then divided by the sum of all importances to get the normalized values.
- Rainfall has the largest importance with a values of 0.16%, while rainfall anomaly is the least important with a value of 0.004%
- These features overall contribute about 0.25% to the final prediction.

# Evaluation on 100 Time Series: SARIMA/SARIMAX

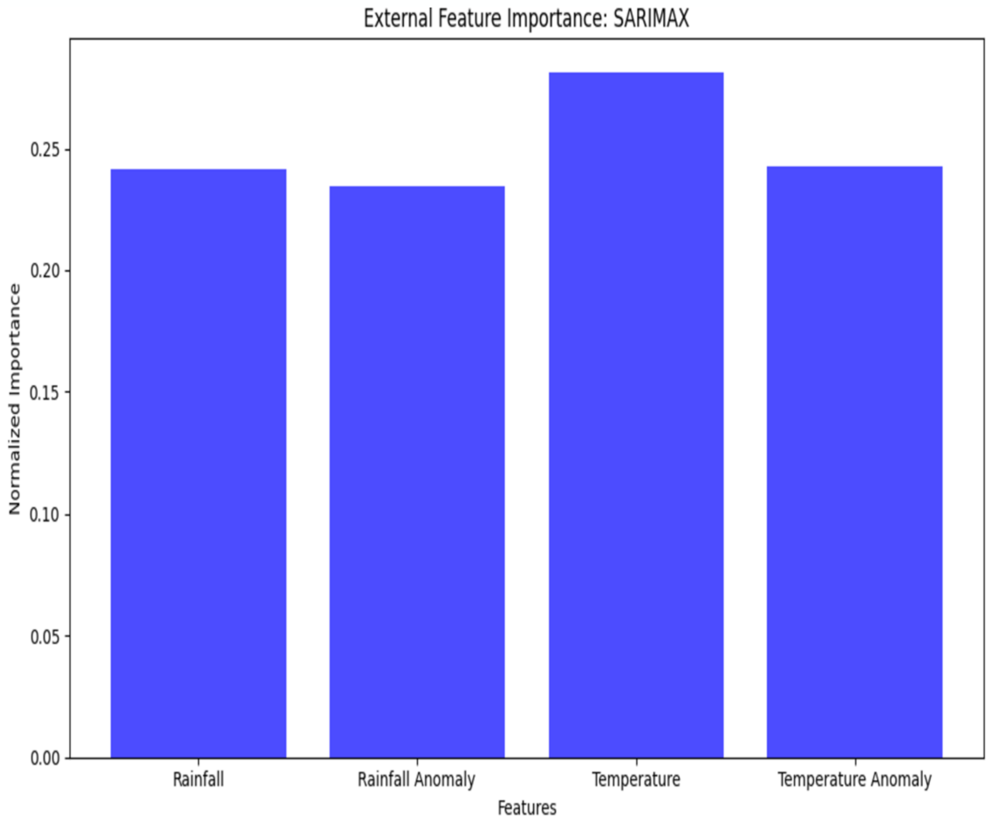
## Moving towards a more realistic scenario



# External Feature Importance: SARIMA/SARIMAX

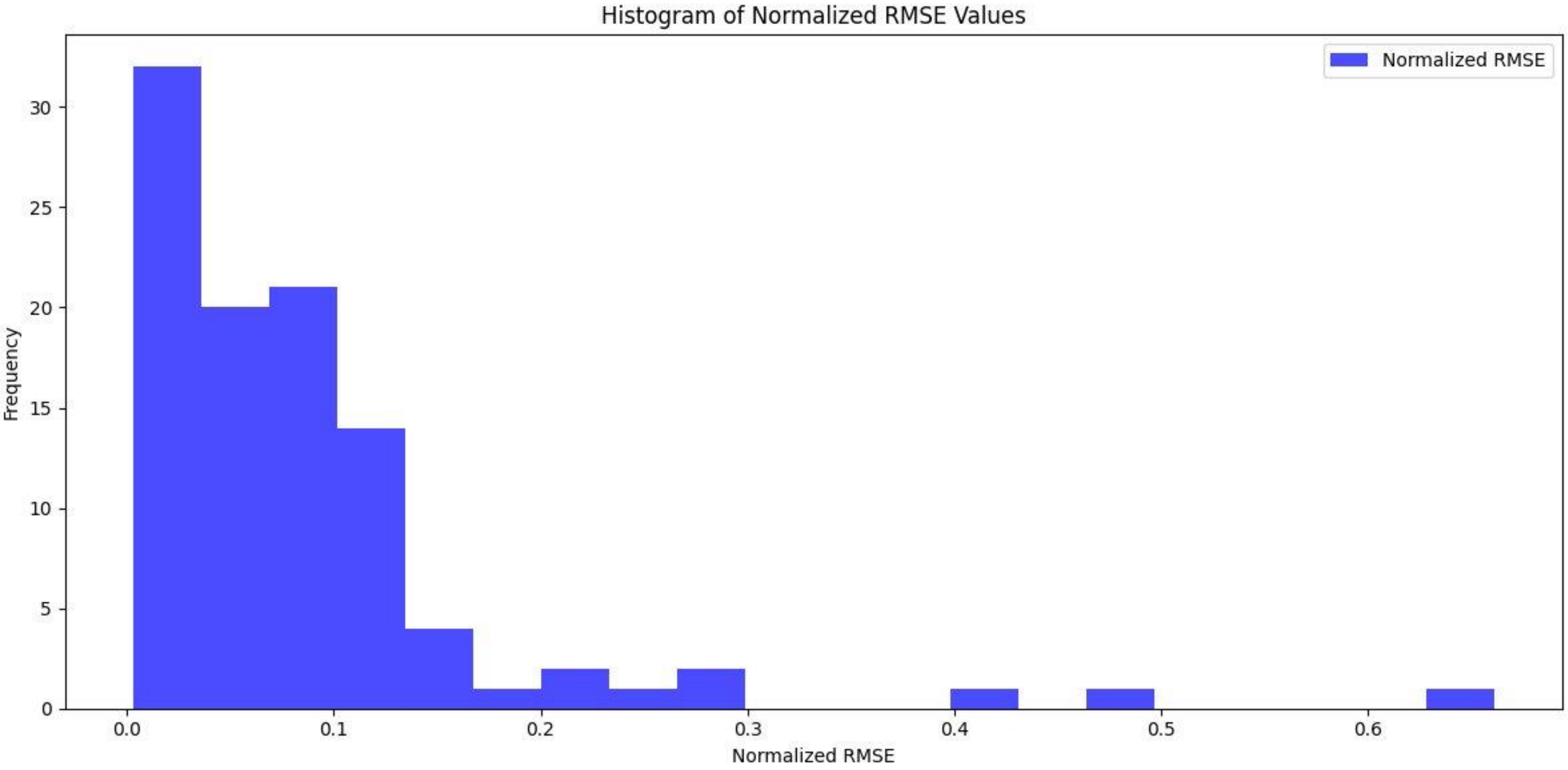
## Moving towards a more realistic scenario




- Temperature shows the highest normalized importance among the four features. This indicates that variations in temperature have the most significant impact on the demand forecasts.
- Rainfall Anomaly and Temperature Anomaly have similar importance levels, suggesting that deviations from long-term averages (anomalies) in both rainfall and temperature contribute moderately to the model's accuracy.
- The results highlight that weather-related features (temperature and rainfall) are critical drivers in the SARIMAX model, with temperature contributing the most.

# *Evaluation on 100 Time Series: Neural Networks*

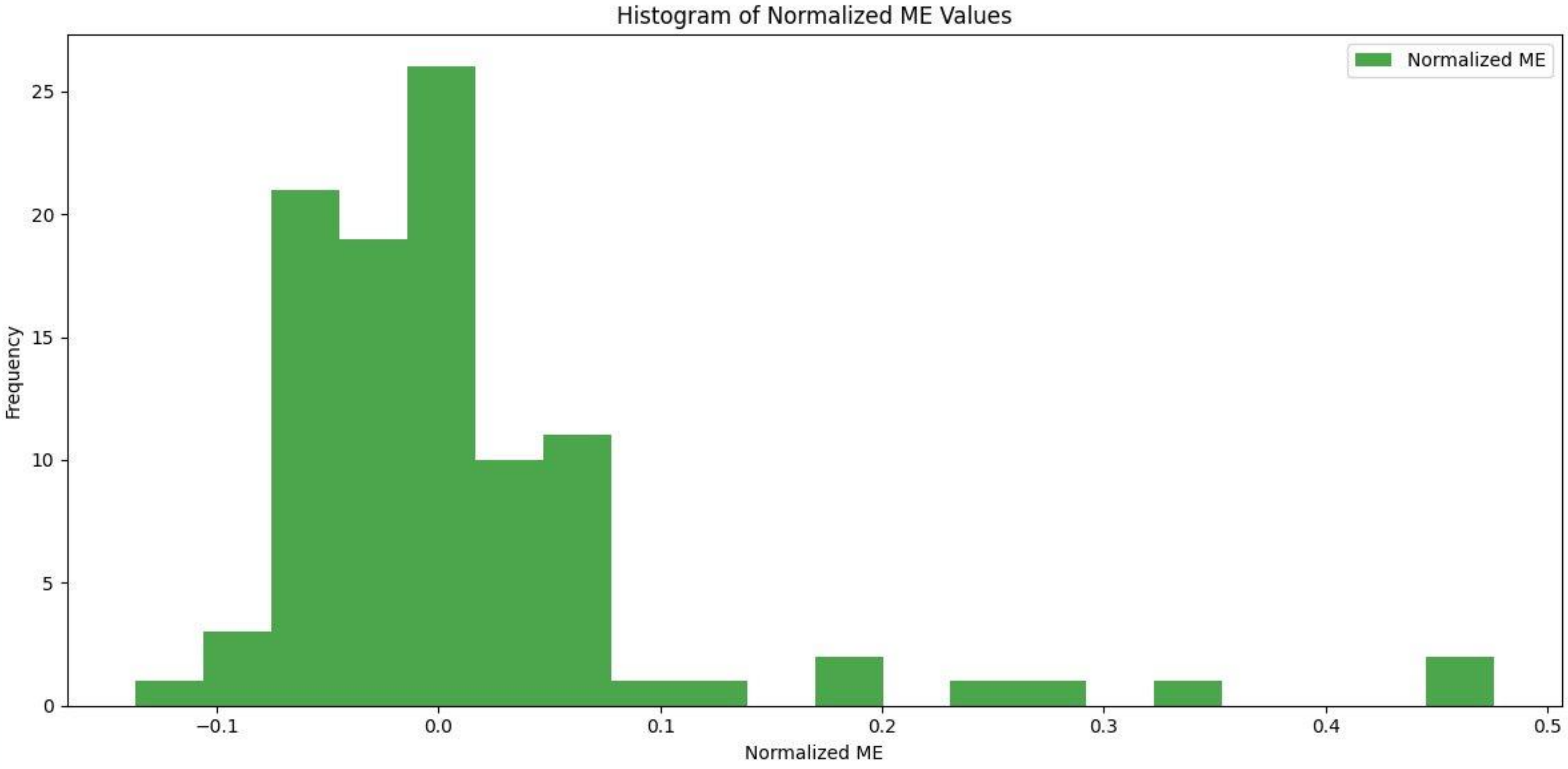
## Moving towards a more realistic scenario




- 100 6 month validation sets
- RMSE normalized by: RMSE / average demand

# *Neural Network ME*

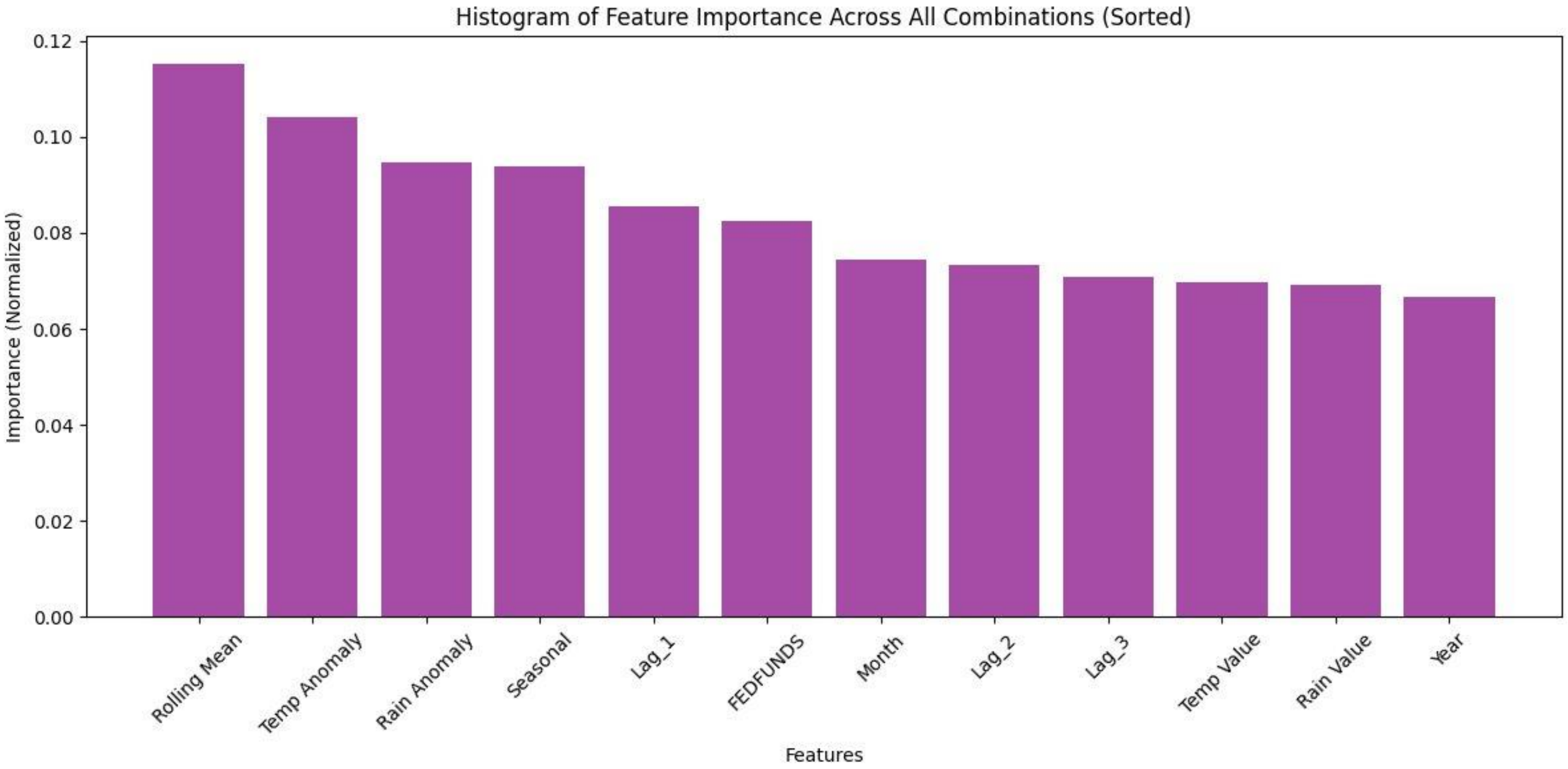
## Moving towards a more realistic scenario




- ME normalized by: ME / average demand

# External Feature Importance: Neural Network

## Moving towards a more realistic scenario




- Feature importance normalized by dividing each feature by the total sum
- 2 Month rolling mean is the most important

# *Spring 2025 Goals*

**What's next?**

## To-be Added based on progress in last 2 weeks:

Expand to Probabilistic Forecasting:

- Quantile Regression
- Bayesian Structural Time Series
- Prophet with Uncertainty Intervals
- Probabilistic SARIMA (SARIMA with prediction intervals)
- Gaussian Processes for Time Series Forecasting
- Monte Carlo Simulations for Demand Forecasting

Classification and Model Application for 1.6 Million Timeseries:

- Classify into buckets (Smooth, Lumpy, Intermittent)
- Find best model for each classified bucket data

# End Objective (Ananya & raj)

- **Quantile Regression** - Predict quantiles with accurate uncertainty estimates.
- **Better than point forecasting** - Range of possible outcomes instead of a single value, QR handles asymmetric data distributions, highlights how external variables impact demand variability. Ideal for risk assessment and inventory planning under uncertainty
- **Bayesian Structural Time Series -** offers several advantages over point forecasting by leveraging its probabilistic nature and flexible structure
- **Better than point forecasting -** full posterior distributions for predictions, Incorporates trend, seasonality, and external regressors, Causal Analysis and Impact Measurement.

- **SARIMA –** Handles seasonality and trends, improving forecast accuracy.
- **Better than point forecasting** - Captures seasonal variability enhancing predictive power and risk assessment. Accounts for periodic fluctuations, providing a comprehensive view of future trends
- **Gaussian Process -** Offers full posterior distributions for predictions, incorporating non-linearity and external regressions
- **Better than point forecasting –** Provides probabilistic forecasts with uncertainity estimates, ideal for risk assessment. Handles non-linear relationships and complex patterns, incorporating trends, seasonality, and external factors.

# *Extending this Semester's Work (Gopal & Logan)*

## What's Next?

### Exploring Deep Learning & Foundational Model

- Transition from point forecasting to **probabilistic forecasting**.
- Implement advanced techniques:
- **Deep Learning Models:** LSTM with probabilistic loss functions (e.g., Gaussian Likelihood).
- **Foundational Models for Time Series:**
    - TimeGPT
    - MOIRAI
    - Lag-LLaMA
    - Enhance accuracy and capture uncertainty for demand predictions.

### Expected Outputs

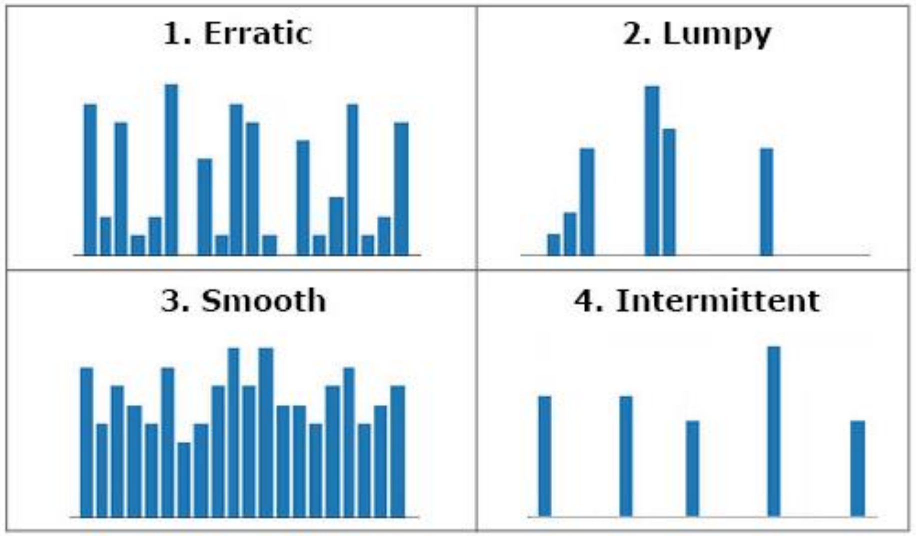
- **Prediction Intervals:** 95% Confidence Intervals for demand forecasts.
- **Probability Distributions:** A distribution of possible outcomes for demand, showing the likelihood of different demand values.
- **Probability Ranges:** Probabilistic outcomes to support better inventory management.

# *Roadmap for Testing and Comparison 1 (frank and sahil)*

**Moving towards a more realistic scenario**

How to test all 1.6 million time series

1. Split them into buckets (smooth, bumpy, intermittent)
2. Our goal: find best model for each type of bucket




The models we are considering:

- Quantile Regression
- Bayesian Structural Time Series
- Prophet with Uncertainty Intervals
- Probabilistic SARIMA (SARIMA with prediction intervals)
- Gaussian Processes for Time Series Forecasting
- Monte Carlo Simulations for Demand Forecasting

# *Roadmap for Testing and Comparison 2 (frank and sahil)*

**Moving towards a more realistic scenario**

Roadmap for testing and compare models (for single time series):

1. Prepare data – Follow the data preprocessing steps from this semester
   - 80% training and 20% testing data split
2. Use all models listed in the previous slide
3. Evaluate results with metrics
   - Accuracy metrics – Mean absolute error (MAE), root mean absolute error (RMSE)
   - Probabilistic metrics – Likelihood ratio test (compare 2 likelihoods of correct prediction)
4. Compare results between models, select best ones
   - Put all models in a table w/ values for each metric
   - One who has the largest amount of most accurate metrics will be selected for time series

# Thank You

We would like to thank our mentors, Yarong Chen & Gonzalo Martinez from John Deere for their constant guidance and support throughout the semester.

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